

Asset Allocation - Portfolio Analysis

ELE Group 3

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Q1 Returns Statistics

The objective of this analysis is to examine the statistical properties of daily equity index returns across major international markets and evaluate whether it is reasonable to approximate the return process by an independent and identically distributed (i.i.d.) Gaussian model.

The analysis focuses on distributional characteristics of returns, stock dependence, and cross-market dependence. The dataset consists of daily price levels for eight equity indices: TSX (Canada), CAC (France), DAX (Germany), Eurostoxx 50, Nikkei 225 (Japan), FTSE 100 (UK), S&P 500 (US), and IBOVESPA (Brazil). All indices are analysed in local currency terms to isolate equity market behavior from exchange rate effects and to ensure comparability across markets.

Although the dataset contains exchange rate information, local-currency analysis is appropriate here. Converting equity indices into USD incorporate currency risk into the return process, thereby increasing volatility, altering skewness and kurtosis, strengthening cross-market correlations due to a common currency factor.

Data Cleaning and Processing

	tsx	cac	dax	eurostoxx50	nikkei225	ftse100	sp500	ibovespa
date								
2007-03-30	13165.500000	5634.160156	6917.029785	4181.029785	17287.650391	6308.03	1420.86	45805
2007-04-02	13265.799805	5645.560059	6937.169922	4189.549805	17028.410156	6315.53	1424.55	45597
2007-04-03	13361.200195	5711.910156	7045.560059	4246.299805	17244.050781	6366.11	1437.77	46288
2007-04-04	13448.299805	5739.009766	7073.910156	4261.830078	17544.089844	6364.70	1439.37	46554
2007-04-05	13425.000000	5741.379883	7099.910156	4271.540039	17491.419922	6397.34	1443.76	46647
...	...	...	...	...	...	...	...	...
2021-12-21	20924.900391	6964.990234	15447.440430	4174.990234	28517.589844	7297.41	4649.23	105500
2021-12-22	21070.099609	7051.669922	15593.469727	4217.060059	28562.210938	7341.66	4696.56	105244
2021-12-23	21218.900391	7106.149902	15756.309570	4265.859863	28798.369141	7373.34	4725.79	104891
2021-12-29	21344.699219	7161.520020	15852.250000	4284.830078	28906.880859	7420.69	4793.06	104107
2021-12-30	21294.599609	7173.229980	15884.860352	4306.069824	28791.710938	7403.01	4778.73	104822

Return Construction

We calculate the daily log returns using the formula:

$$r_{i,t} = \ln\left(\frac{R_{i,t}}{R_{i,t-1}}\right)$$

Log returns are preferred because they are time-additive and approximately symmetric for gains and losses, which enables statistical analysis and aggregation across time. Prior to analysis, the return series are verified to contain no infinite or missing values, ensuring all statistics are computed on a consistent dataset.

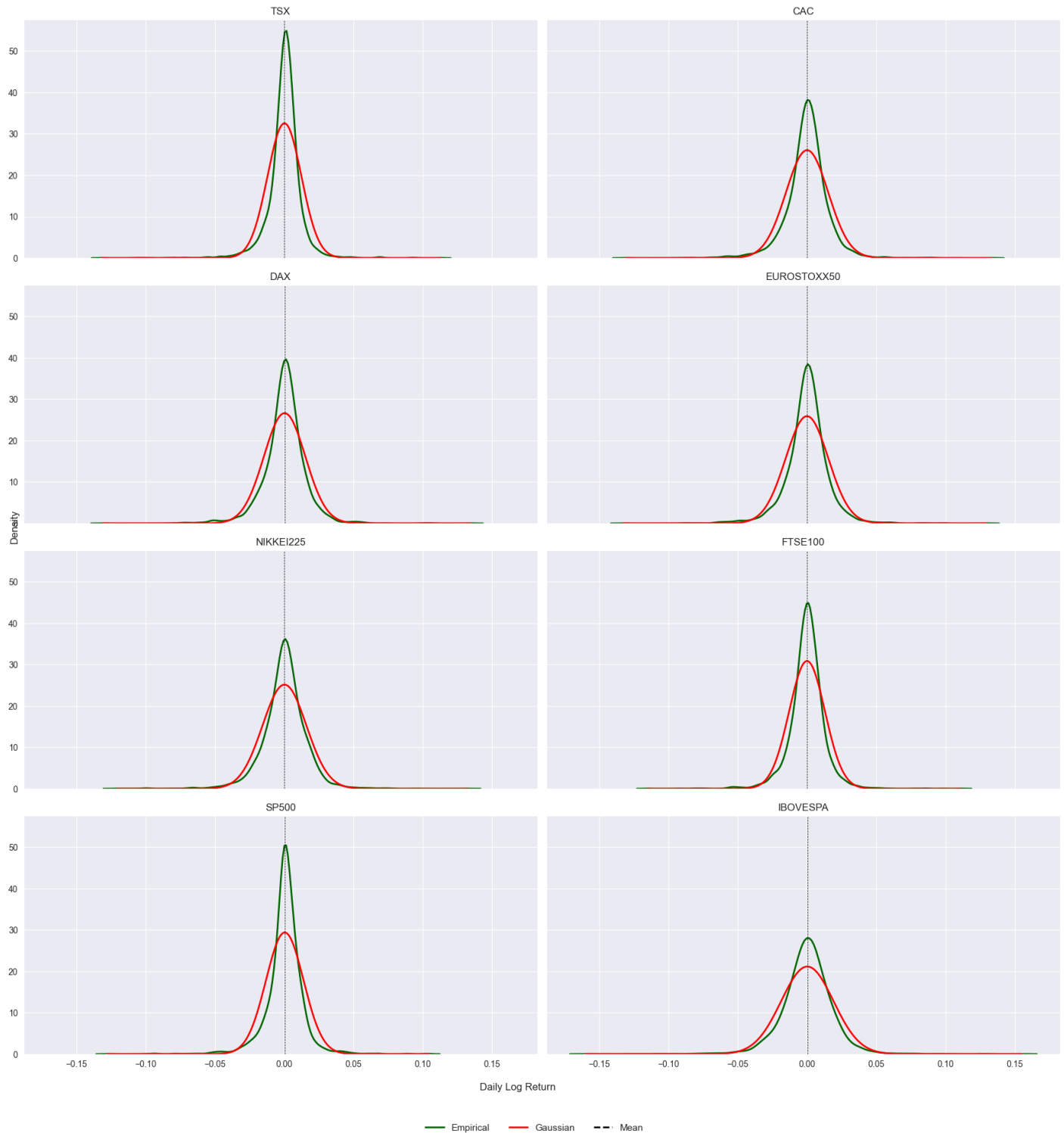
date	tsx	cac	dax	eurostoxx50	nikkei225	ftse100	sp500	ibovespa
2007-04-02	0.007590	0.002021	0.002907	0.002036	-0.015109	0.001188	0.002594	-0.004551
2007-04-03	0.007166	0.011684	0.015504	0.013455	0.012584	0.007977	0.009237	0.015041
2007-04-04	0.006498	0.004733	0.004016	0.003651	0.017250	-0.000222	0.001112	0.005730
2007-04-05	-0.001734	0.000413	0.003669	0.002276	-0.003007	0.005115	0.003045	0.001996
2007-04-10	0.000469	0.004326	0.009359	0.006982	0.009857	0.003188	0.003202	0.011234
...	...	...	...	...	...	...	...	...
2021-12-21	0.018653	0.013718	0.013541	0.016388	0.020540	0.013712	0.017622	0.004560
2021-12-22	0.006915	0.012368	0.009409	0.010026	0.001563	0.006045	0.010129	-0.002429
2021-12-23	0.007037	0.007696	0.010389	0.011506	0.008234	0.004306	0.006204	-0.003360
2021-12-29	0.005911	0.007762	0.006071	0.004437	0.003761	0.006401	0.014134	-0.007502
2021-12-30	-0.002350	0.001634	0.002055	0.004945	-0.003992	-0.002385	-0.002994	0.006844

Plotting Empirical vs Gaussian Daily Log Returns Distributions

Across all indices, empirical return distributions deviate clearly from the Gaussian benchmark. The empirical densities are more sharply peaked and exhibit heavier tails, suggesting frequent small returns and a higher likelihood of extreme outcomes than predicted by a normal distribution.

Moreover, the left tail is typically heavier than the right, consistent with negative skewness and greater downside risk. While the Gaussian model approximates the centre of distribution, it underestimates tail risk.

Empirical vs Gaussian Daily Return Distributions

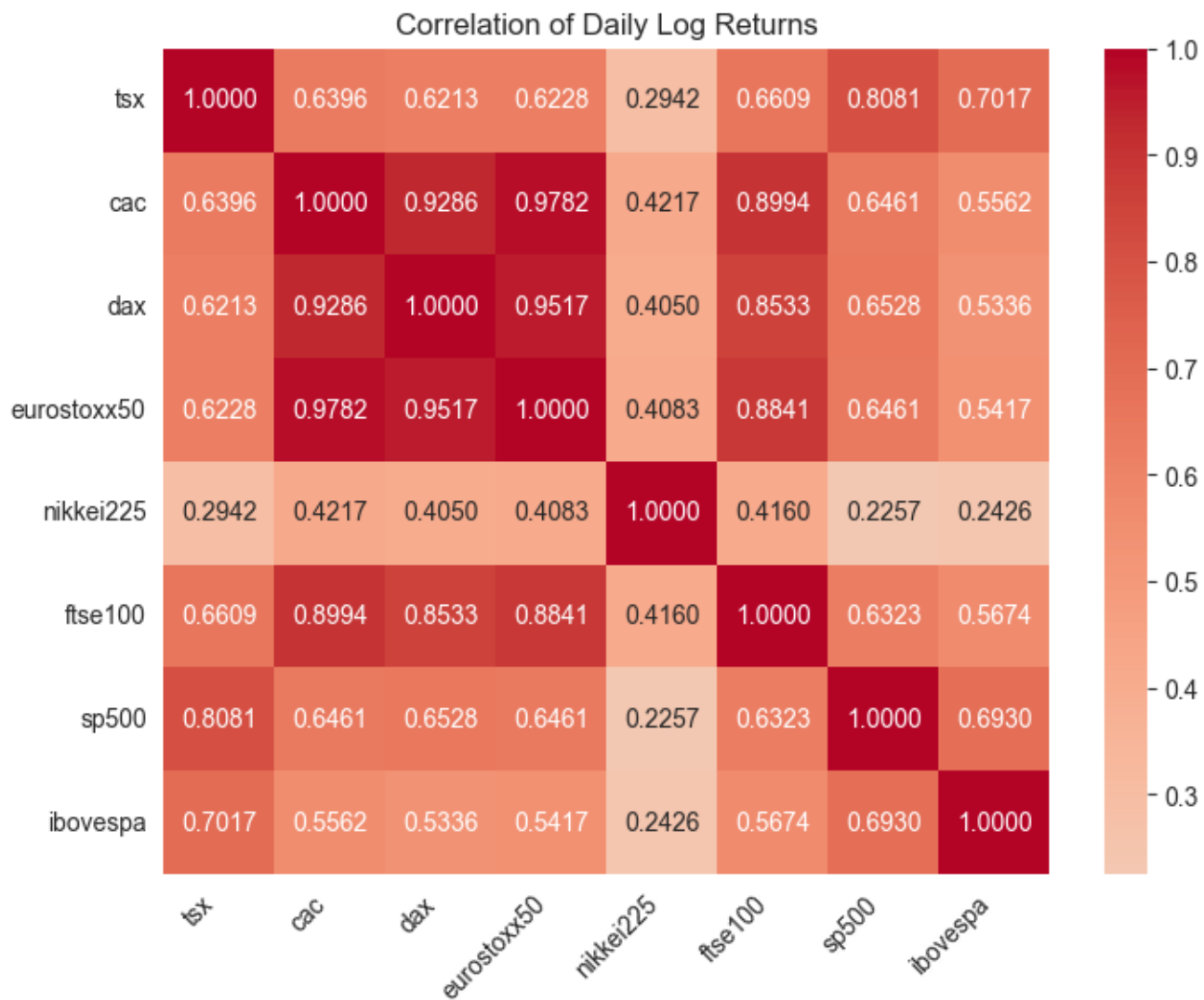


	tsx	cac	dax	eurostoxx50	nikkei225	ftse100	sp500	ibovespa
count	3240.000000	3240.000000	3240.000000	3240.000000	3240.000000	3240.000000	3240.000000	3240.000000
mean	0.000148	0.000075	0.000257	0.000009	0.000157	0.000049	0.000374	0.000256
std	0.012258	0.015316	0.014979	0.015405	0.015865	0.012931	0.013581	0.018863
min	-0.131758	-0.130983	-0.130549	-0.132405	-0.121110	-0.115124	-0.127652	-0.159930
25%	-0.004112	-0.006507	-0.006167	-0.006469	-0.007130	-0.005497	-0.004264	-0.009018
50%	0.000836	0.000598	0.000785	0.000489	0.000560	0.000389	0.000812	0.000549
75%	0.005430	0.007418	0.007439	0.007126	0.008037	0.006291	0.006232	0.009978
max	0.112945	0.133048	0.134627	0.129509	0.132346	0.111124	0.104236	0.154728

Correlation matrix

The correlation matrix reveals cross-market co-movement, particularly among European indices such as CAC, DAX, and Eurostoxx 50. This shows shared regional economic conditions and global risk factors.

In contrast, the Nikkei 225 exhibits relatively lower correlations with other markets, suggesting some diversification potential. Moreover, correlations are generally positive and economically significant, indicating that returns are not independent across assets. This cross-sectional dependence violates the independence assumption and has direct implications for international diversification.



Metrics

Mean Returns (Rewards)

The mean daily returns provide a measure of the average reward earned by holding each index. Across markets, average daily returns are small, consistent with most equity market gains that accrue slowly over time rather than through large daily movements. Difference in mean returns across indices reflect heterogeneity in long-run performance across regions.

However, the mean provides limited insight into risk and is insufficient for evaluating return distributions.

Mean (average daily return):

tsx	0.000148
cac	0.000075
dax	0.000257
eurostoxx50	0.000009
nikkei225	0.000157
ftse100	0.000049
sp500	0.000374
ibovespa	0.000256

dtype: float64

Standard Deviation (Volatility by Index)

The standard deviation of daily returns captures return volatility, which is a measure of risk. Volatility varies significantly across indices, with emerging markets like IBOVESPA displaying higher volatility relative to developed markets.

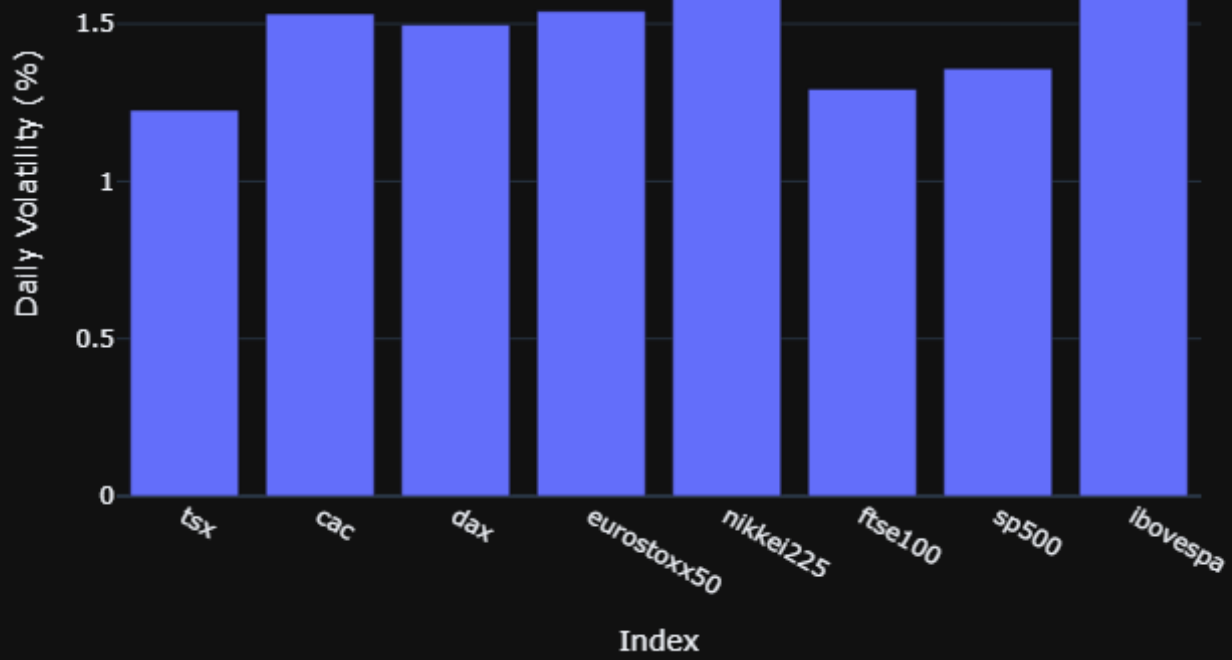
This reflects differences in economic stability, and exposure to global shocks. However, volatility alone does not capture asymmetry or tail risk, showing the need for further analysis.

Standard deviation (daily volatility):

tsx	0.012258
cac	0.015316
dax	0.014979
eurostoxx50	0.015405
nikkei225	0.015865
ftse100	0.012931
sp500	0.013581
ibovespa	0.018863

dtype: float64

Daily Volatility of Equity Index Returns



Skewness (Asymmetric Risk)

All indices in the sample dataset exhibit negative skewness. This indicates that large negative returns occur more frequently, or are more severe than the large positive returns. The downside risk dominates upside surprises. From a risk perspective, negative skewness reflects crash risk, which is not captured by variance alone. The presence of negative skewness directly contradicts the symmetry implied by a Gaussian distribution.

Kurtosis (Tail Risk)

Kurtosis values are above zero for all indices, indicating excess kurtosis relative to gaussian distribution. This implies that return distributions are fat-tailed, with extreme outcomes occurring more frequently than predicted by Gaussian models.

High kurtosis demonstrates extreme losses are more likely than standard models. Together with negative skewness, excess kurtosis provides strong evidence against normality.

Skewness:

tsx	-0.941299
cac	-0.158966
dax	-0.118485
eurostoxx50	-0.227512
nikkei225	-0.419489
ftse100	-0.242695
sp500	-0.637743
ibovespa	-0.292861

dtype: float64

Kurtosis:

tsx	18.396061
cac	8.578744
dax	8.565136
eurostoxx50	7.929156
nikkei225	7.536117
ftse100	10.182191
sp500	12.021852
ibovespa	9.314212

dtype: float64

Skewness vs Excess Kurtosis of Daily Returns



Autocorrelation (Lag 1)

First order autocorrelation of daily returns is close to zero across all indices. This suggests that returns exhibit little linear dependence over time, consistent with weak-form market efficiency i.e. daily returns provide limited information about future returns. It supports the independence over time of the i.i.d assumption for the daily frequency.

First-order autocorrelation:

```
tsx          -0.098100
cac          -0.029844
dax          0.003019
eurostoxx50 -0.033778
nikkei225    -0.035715
ftse100      -0.045531
sp500        -0.154566
ibovespa     -0.077461
dtype: float64
```

Summarising statistics

	Mean	Standard Deviation	Skewness	Kurtosis	Autocorrelation
tsx	0.000148	0.012258	-0.941299	18.396061	-0.098100
cac	0.000075	0.015316	-0.158966	8.578744	-0.029844
dax	0.000257	0.014979	-0.118485	8.565136	0.003019
eurostoxx50	0.000009	0.015405	-0.227512	7.929156	-0.033778
nikkei225	0.000157	0.015865	-0.419489	7.536117	-0.035715
ftse100	0.000049	0.012931	-0.242695	10.182191	-0.045531
sp500	0.000374	0.013581	-0.637743	12.021852	-0.154566
ibovespa	0.000256	0.018863	-0.292861	9.314212	-0.077461

Analysis

The empirical evidence helps conclude that while daily equity returns demonstrate weak serial dependence, consistent with approximate independence over time, their distributional properties deviate from Gaussianity. All indices display negative skewness and excess kurtosis, and heavy tails indicate frequent extreme events that are not captured by a normal distribution.

Additionally, the correlation analysis reveals economically significant cross-market dependence, particularly among European equity indices, reflecting common regional and global risk factors. This cross-market dependence violates the independence assumption.

Lastly, although the i.i.d. Gaussian model provides a convenient analytical benchmark, it does not offer a realistic description of the empirical equity return process. Equity returns are nearly unpredictable on a daily basis, yet they are characterised by asymmetric risk, fat tails, and strong cross-market co-movement.

Q2 Mean–Variance Analysis

Part 1: Unconstrained version (short selling allowed)

We compute Q2 in two variants:

- 1) Local-currency index returns (no FX conversion).
- 2) USD-denominated returns ($\text{index} \times \text{FX}$), using the FX series provided.

For both variants:

- Annualize mean returns and covariance.
- Compute the unconstrained mean–variance efficient frontier.
- Compute and mark the tangency portfolio (maximum Sharpe). Since r_f is not provided, set $r_f = 0$.
- Print tangency weights (extreme weights are expected when short-selling is allowed).

Preliminary Analysis

Cleaning the Data

The first two rows of the Excel sheet contain metadata (labels and index tickers), while the actual time-series observations begin at row index 2. Accordingly, extract the index tickers from row 0 and construct the price level panel using data from row 2 onward

Store index levels in a single date-indexed DataFrame to standardize downstream operations (log returns, annualization, covariance)

All index levels are strictly positive (safe for log returns).

Min/Max level by index (quick plausibility check):

	min	max
SPTSX INDEX	7566.899902	21768.500000
CAC INDEX	2519.290039	7173.229980
DAX INDEX	3666.409912	16251.129883
SX5E INDEX	1809.979980	4556.970215
NKY INDEX	7054.979980	30670.099609
UKX INDEX	3512.090000	7877.450000
SPX INDEX	676.530000	4793.060000
IBOV INDEX	29435.000000	130776.000000

FX table shape: (3241, 6)

Date range: 2007-03-30 00:00:00 to 2021-12-30 00:00:00

Total missing values in FX table: 0

First 5 rows of FX (USD per local currency unit):

	CAD.1	EUR.3	JPY.1	GBP.1	USD.1	BRL.1
date						
2007-03-30	0.86633	1.33590	0.00849	1.95252	1.0	0.48662
2007-04-02	0.86490	1.33701	0.00849	1.97820	1.0	0.48936
2007-04-03	0.86315	1.33250	0.00841	1.97340	1.0	0.49167
2007-04-04	0.86304	1.33740	0.00842	1.97640	1.0	0.49293
2007-04-05	0.86888	1.34320	0.00842	1.97149	1.0	0.49324

Quick plausibility check (typical magnitudes):

	CAD.1	EUR.3	JPY.1	GBP.1	USD.1	BRL.1
min	0.686662	1.039047	0.007960	1.159555	1.0	0.169883
mean	0.866398	1.257054	0.009905	1.512662	1.0	0.397014
max	1.088495	1.598798	0.013203	2.108192	1.0	0.652018

This explicit mapping avoids accidental currency mismatch. Note: multiple Euro-area indices share the same EURUSD series

FX mapping (ticker -> FX column):

```
sptsx    -> CAD.1
cac      -> EUR.3
dax      -> EUR.3
sx5e     -> EUR.3
nky      -> JPY.1
ukx      -> GBP.1
spx      -> USD.1
ibov     -> BRL.1
```

Verify the conversion by checking that P^{USD}/P^{LC} equals the FX series used for that asset. This confirms both (i) the FX interpretation and (ii) correct ticker-to-currency mapping

USD price table shape: (3241, 8)

Total missing values in USD price table: 0

First 5 rows of USD-denominated index levels:

	sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
date								
2007-03-30	11405.61	7526.66	9240.45	5585.43	146.74	12316.52	1420.86	22289.54
2007-04-02	11473.62	7548.15	9275.04	5601.45	144.61	12493.38	1424.55	22313.21
2007-04-03	11532.69	7611.11	9388.19	5658.19	145.07	12562.88	1437.77	22758.24
2007-04-04	11606.37	7675.35	9460.64	5699.77	147.72	12579.21	1439.37	22947.68
2007-04-05	11664.78	7711.83	9536.61	5737.54	147.37	12612.30	1443.76	23008.31

['sptsx', 'cac', 'dax', 'sx5e', 'nky', 'ukx', 'spx', 'ibov']

Conversion identity check (USD/Local should match FX):

	date	ticker	USD/Local ratio	FX used	difference
0	2007-03-30	sptsx	0.866326	0.866326	0.0
1	2007-03-30	nky	0.008488	0.008488	0.0
2	2007-03-30	ukx	1.952515	1.952515	0.0
3	2009-07-22	sptsx	0.908843	0.908843	-0.0
4	2009-07-22	nky	0.010687	0.010687	0.0
5	2009-07-22	ukx	1.645007	1.645007	0.0
6	2011-11-04	sptsx	0.992142	0.992142	0.0
7	2011-11-04	nky	0.012812	0.012812	0.0
8	2011-11-04	ukx	1.603721	1.603721	0.0

Daily log returns (local) shape: (3240, 8)

Daily log returns (USD) shape: (3240, 8)

Missing values (local returns): 0

Missing values (USD returns): 0

Daily return summary (LOCAL):

	mean	std	min	max
sptsx	0.000148	0.012258	-0.131758	0.112945
cac	0.000075	0.015316	-0.130983	0.133048
dax	0.000257	0.014979	-0.130549	0.134627
sx5e	0.000009	0.015405	-0.132405	0.129509
nky	0.000157	0.015865	-0.121110	0.132346
ukx	0.000049	0.012931	-0.115124	0.111124
spx	0.000374	0.013581	-0.127652	0.104236
ibov	0.000256	0.018863	-0.159930	0.154728

Daily return summary (USD):

	mean	std	min	max
sptsx	0.000117	0.015251	-0.134755	0.111736
cac	0.000025	0.018352	-0.135855	0.242904
dax	0.000207	0.018058	-0.135420	0.233184
sx5e	-0.000041	0.018481	-0.137276	0.243568
nky	0.000165	0.016163	-0.180166	0.176489
ukx	-0.000065	0.015374	-0.122238	0.133055
spx	0.000374	0.013581	-0.127652	0.104236
ibov	-0.000060	0.024794	-0.196712	0.258927

Annualisation

Annualization assumes independent daily returns so mean scales linearly with time and variance scales linearly with time. These are direct inputs to Markowitz mean-variance optimization.

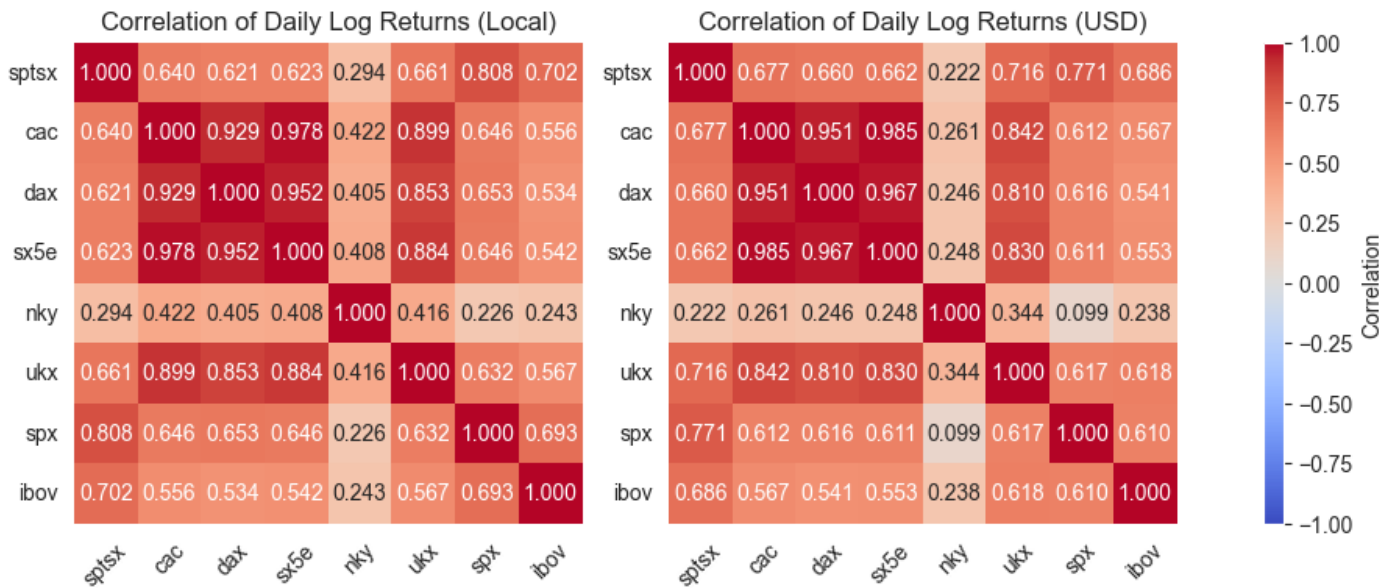
Annualized inputs (LOCAL):

	mu_ann	vol_ann
sptsx	0.037400	0.194583
cac	0.018784	0.243140
dax	0.064663	0.237791
sx5e	0.002292	0.244546
nky	0.039674	0.251847
ukx	0.012449	0.205276
spx	0.094338	0.215596
ibov	0.064390	0.299445

Annualized inputs (USD):

	mu_ann	vol_ann
sptsx	0.029466	0.242104
cac	0.006178	0.291322
dax	0.052057	0.286656
sx5e	-0.010314	0.293384
nky	0.041593	0.256576
ukx	-0.016259	0.244062
spx	0.094338	0.215596
ibov	-0.014995	0.393585

Correlation of USD vs Local Log Returns



Correlation Structure Analysis

The correlation matrices reveal important patterns in the co-movement of international equity markets, both in local currency and USD-denominated terms.

Local Currency Correlations: In local currency terms, the most striking feature is the extremely high correlation among European equity indices. The CAC, DAX, and SX5E exhibit correlations exceeding 0.92 with each other, and all three show correlations above 0.84 with the UKX (FTSE 100). This near-perfect co-movement reflects the integrated nature of European equity markets, shared monetary policy (for Eurozone members), and common exposure to regional economic cycles. In contrast, the Nikkei 225 shows substantially lower correlations with all other indices (ranging from 0.29 to 0.42), indicating that Japanese equities provide meaningful diversification benefits. The S&P 500 occupies a middle ground, with moderate-to-high correlations with developed markets (0.63-0.81) but lower correlation with the Nikkei (0.23).

USD-Denominated Correlations: When returns are converted to USD terms, several important changes emerge. First, correlations among European indices remain extremely high (0.95-0.99), suggesting that currency conversion does little to reduce their co-movement—the EUR/USD exchange rate affects all European indices similarly. Second, and more significantly, the Nikkei's correlations with other markets decrease substantially when measured in USD (dropping to 0.10-0.26 range vs. SPX and European markets). This occurs because the JPY/USD exchange rate introduces an additional source of variation that is largely uncorrelated with equity returns in other markets. Third, correlations between North American indices (SPX, TSX) and European indices decrease modestly (from ~0.62-0.66 to ~0.61-0.68 for SPX), reflecting the diversification benefit of currency exposure.

Implications for Portfolio Construction: These correlation patterns explain why the mean-variance optimization concentrates in SPX and NKY while excluding European indices. The high correlation among European markets (0.95+) means that holding multiple European indices provides minimal marginal diversification benefit—essentially, CAC, DAX, SX5E, and UKX behave as near-perfect substitutes. Meanwhile, the Nikkei's low correlation with SPX (0.10 in USD terms) makes it an ideal diversification partner, allowing the portfolio to reduce risk through exposure to relatively independent market movements. The optimization effectively recognizes that diversification quality (low correlation) matters more than diversification quantity (number of assets held), leading to the concentrated two-asset solution that dominates the constrained efficient frontier.

Part 1: Unconstrained (Short Selling Allowed)

Tangency Portfolio (Maximum Sharpe, $r_f = 0$)

With $r_f = 0$, the tangency portfolio solves:

$$\max_w \frac{\mu^\top w}{\sqrt{w^\top \Sigma w}} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1.$$

In the unconstrained case, the tangency direction is proportional to $\Sigma^{-1}\mu$. We compute:

$$\tilde{w} = \Sigma^{-1}\mu, \quad w = \frac{\tilde{w}}{\mathbf{1}^\top \tilde{w}},$$

which enforces the budget constraint $\mathbf{1}^\top w = 1$. Extreme portfolio weights may arise because short-selling and leverage are permitted in the unconstrained setting.

Results: Efficient Frontier and Tangency Portfolio ($r_f = 0$)

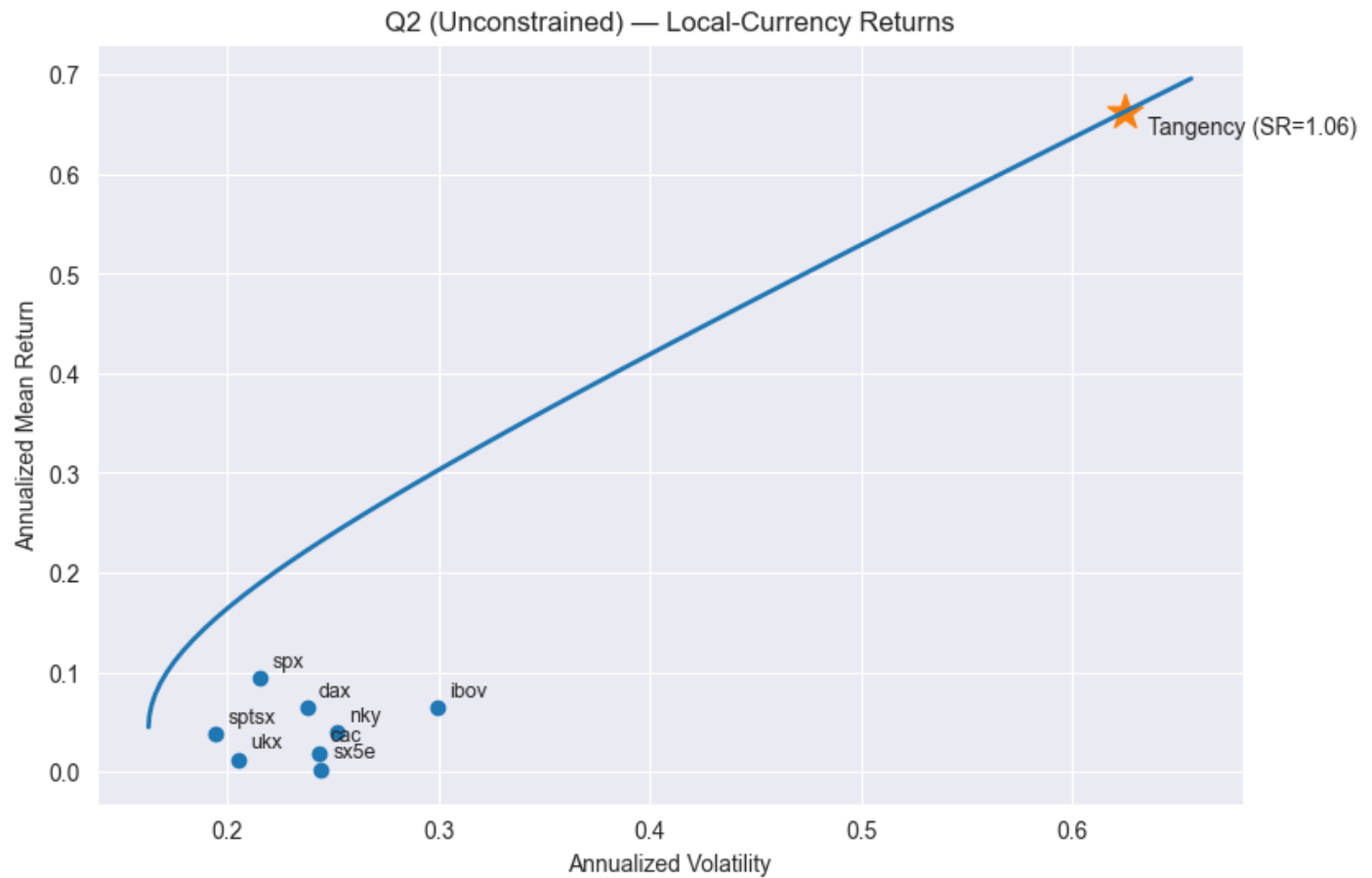
For each data variant (local-currency and USD-denominated), we proceed as follows: 1. Compute annualized μ and Σ . 2. Compute the unconstrained efficient frontier via quadratic programming. 3. Compute the tangency portfolio (maximum Sharpe ratio with $r_f = 0$). 4. Plot individual assets together with the efficient frontier and report tangency portfolio weights (extreme weights are expected under short-selling).

Tangency checks:

```
sum(w) = 1.0
mu_tan = 0.6628056039239415
vol_tan = 0.6251403584675704
SR (rf=0) = 1.060250861980342
```

Frontier constraint checks:

```
max |sum(w) - 1| = 1.1102230246251565e-15
max |mu(w) - r| = 2.220446049250313e-16
```



Tangency weights (sorted):

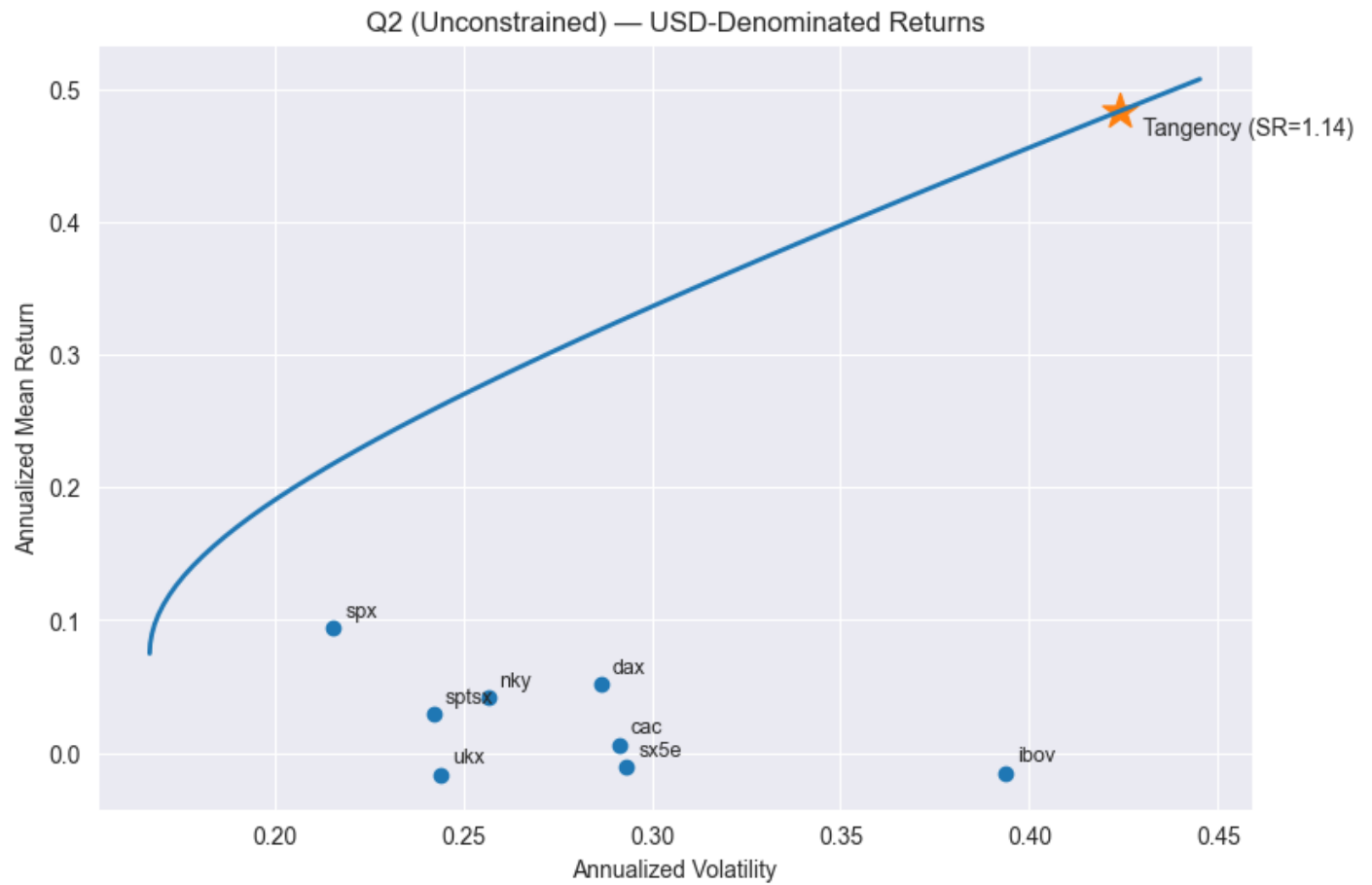
```
dax      6.504534
cac      4.550528
spx      2.416217
nky      0.379206
ibov     -0.010319
ukx      -0.732950
sptsx    -1.387717
sx5e     -10.719499
dtype: float64
```

Tangency checks:

```
sum(w) = 0.9999999999999996
mu_tan = 0.4831227298599197
vol_tan = 0.42391297581752735
SR (rf=0) = 1.1396743138805903
```

Frontier constraint checks:

```
max |sum(w) - 1| = 1.3322676295501878e-15
max |mu(w) - r| = 1.1102230246251565e-16
```



Tangency weights (sorted):

```
dax      4.027815
cac      3.078379
spx      1.731239
nky      0.446342
ibov     -0.284320
sptsx    -0.482369
ukx      -0.834722
sx5e     -6.682364
dtype: float64
```

Analysis: Unconstrained Mean–Variance Frontier and Tangency Portfolio ($r_f = 0$)

1) Efficient frontier shape and interpretation

In both variants (local-currency and USD-denominated), the plotted curve represents the unconstrained mean–variance efficient frontier: for each target annualized return r , we solve for the minimum-variance fully invested portfolio. The frontier has the expected “bullet/arc” shape: near the left edge are low-volatility portfolios (close to the global minimum-variance region), and moving up and to the right corresponds to portfolios with higher target returns that require increasingly aggressive long/short exposures. Because short-selling is permitted, the feasible set expands substantially relative to a long-only case, and the efficient frontier extends to very high expected returns and volatility.

2) Numerical validity of the optimization

The diagnostic prints confirm that the optimization is solved to machine precision. For the frontier portfolios, the constraints $\mathbf{1}^\top w = 1$ and $\mu^\top w = r$ are satisfied with maximum absolute errors on the order of 10^{-15} – 10^{-16} . Similarly, the tangency portfolio weights sum to one (up to floating-point rounding). This confirms that the quadratic programming setup and the Markowitz-to-quadprog mapping are implemented correctly.

3) Tangency portfolio (maximum Sharpe ratio with $r_f = 0$)

The tangency portfolio is marked with a star on each plot. By construction (with $r_f = 0$), it maximizes the Sharpe ratio

$$SR = \frac{\mu_p}{\sigma_p}.$$

For the local-currency case, the tangency portfolio has $\mu_{\text{tan}} \approx 0.663$ and $\sigma_{\text{tan}} \approx 0.625$, yielding $SR \approx 1.06$. For the USD-denominated case, $\mu_{\text{tan}} \approx 0.483$ and $\sigma_{\text{tan}} \approx 0.424$, yielding $SR \approx 1.14$. In both cases, the tangency portfolio lies on the efficient frontier, as expected, since it is the mean–variance optimal portfolio among those that maximize return per unit of risk relative to $r_f = 0$.

4) Extreme weights under unconstrained optimization (short-selling allowed)

The printed tangency portfolio weights are highly leveraged, with large positive and negative values. This is a standard outcome of unconstrained mean–variance optimization: the optimizer exploits small differences in expected returns and correlations by taking large offsetting long and short positions, effectively creating leveraged factor exposures. For example, large positive weights in some indices are financed by substantial short positions in others (e.g., sizeable negative exposure to the EuroStoxx50 in both variants). This behavior illustrates a key point emphasized in the assignment: unconstrained Markowitz solutions can be unstable and extreme, motivating the follow-on exercise with a no-short-sale constraint.

5) Local-currency vs USD-denominated results (effect of FX)

The USD-denominated variant differs because USD returns incorporate both equity index movements and currency movements:

$$r_t^{\text{USD}} = r_t^{\text{LC}} + r_t^{\text{FX}}.$$

This alters both the estimated mean vector μ and covariance matrix Σ , and therefore changes the efficient frontier and tangency portfolio weights. In our results, the USD-denominated tangency portfolio achieves a slightly higher Sharpe ratio than the local-currency tangency portfolio, and the optimal weights shift accordingly, reflecting the additional diversification (or risk) introduced by currency fluctuations. The differences between the two sets of weights and tangency statistics demonstrate that currency denomination materially impacts mean-variance optimal allocations.

6) Key takeaway

Overall, Part 1 demonstrates that the efficient frontier and tangency portfolio can be computed cleanly using quadratic programming, allowing short-selling leads to extreme and leveraged allocations, and then converting returns to USD materially changes optimal portfolios due to FX-driven shifts in μ and Σ .

Part 2: Constrained version (short selling not allowed)

We are now solving the mean-variance problem without short-selling allowed (i.e., no nonnegativity constraints). We added a non negativity constraint into the optimization. For a target portfolio expected return r , the Markowitz efficient portfolio solves:

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \mu^\top w = r \quad w \geq 0.$$

This produces the efficient frontier by varying r over a grid of target returns.

Mapping to quadprog.solve_qp

The quadprog solver handles problems of the form:

$$\min_x \frac{1}{2} x^\top G x - a^\top x \quad \text{s.t.} \quad C^\top x \geq b,$$

We map the Markowitz problem as follows: - $x \equiv w$ (portfolio weights) - $G \equiv \Sigma$ (annualized covariance matrix; ridge-stabilized to ensure positive definiteness) - $a \equiv 0$ (no linear term)

Equality constraints: - $\mathbf{1}^\top w = 1$ - $\mu^\top w = r$

Inequality constraints: - $w \geq 0$

Accordingly, we set: - $C = [\mathbf{1}, \mu]$ (shape $n \times 2$) - $b = [1, r]$ - meq = 2

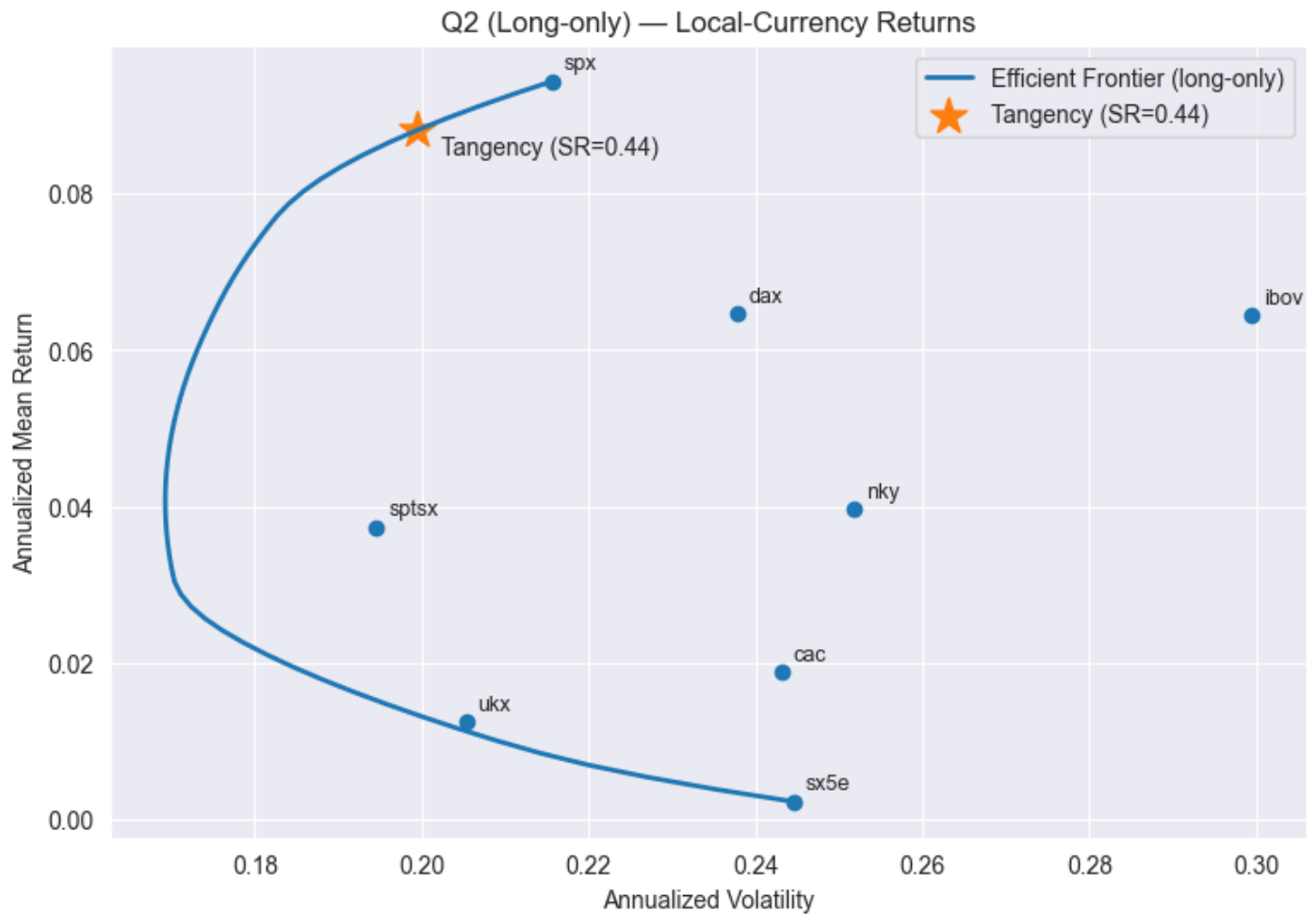
In this problem now Short-selling is not allowed.

Tangency portfolio (LONG-ONLY, rf = 0):

```
sum(w) = 1.0
min(w) = -6.513734074471181e-16
mu_tan = 0.08809638928017097
vol_tan = 0.1994463388611613
SR (rf=0) = 0.4417047200926394
```

Long-only frontier constraint checks:

```
max |sum(w) - 1| = 2.220446049250313e-16
max |mu(w) - r| = 5.551115123125783e-17
min weight (should be >= 0) = -8.793236149593734e-16
```



Tangency weights (sorted):

```
spx      8.858243e-01
nky      1.141757e-01
ukx      1.006240e-18
ibov     0.000000e+00
dax     -1.388850e-17
sptsx   -5.187733e-17
cac     -6.619590e-17
```

```

sx5e    -6.513734e-16
dtype: float64

```

Assets used ($w > 0$):

```

spx    0.885824
nky    0.114176
dtype: float64

```

Tangency portfolio (LONG-ONLY, $r_f = 0$):

```

sum(w) = 1.0
min(w) = -9.50699722448468e-17
mu_tan = 0.08496423951480907
vol_tan = 0.18738602097204554
SR (rf=0) = 0.4534182383193041

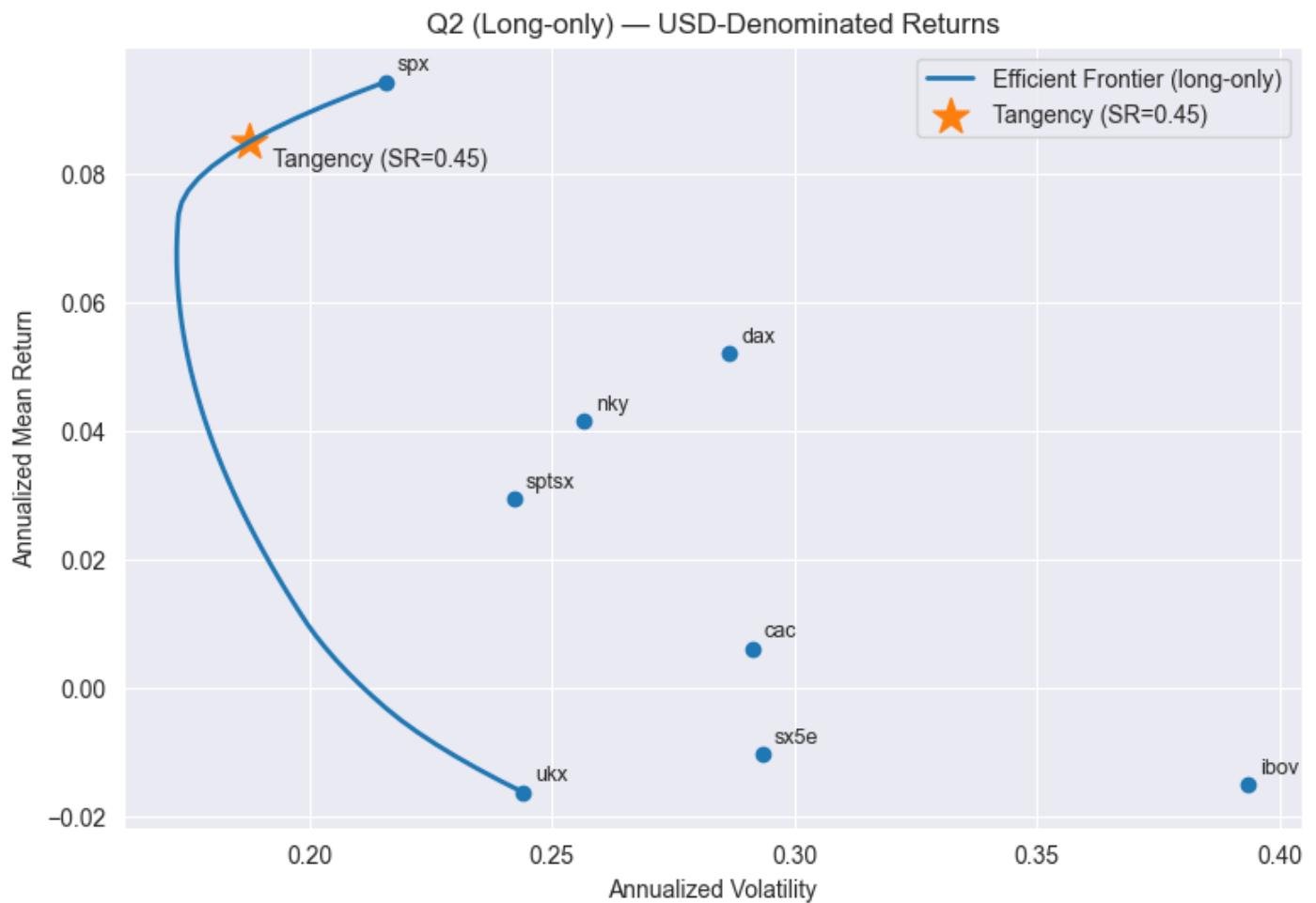
```

Long-only frontier constraint checks:

```

max |sum(w) - 1| = 4.440892098500626e-16
max |mu(w) - r| = 4.9439619065339e-17
min weight (should be >= 0) = -1.7756823950250597e-16

```



Tangency weights (sorted):

```
spx      8.222878e-01
nky      1.777122e-01
sx5e     1.404040e-16
ukx      1.464768e-17
dax      4.471602e-19
sptsx    0.000000e+00
ibov     0.000000e+00
cac      -9.506997e-17
dtype: float64
```

Assets used ($w > 0$):

```
spx      0.822288
nky      0.177712
dtype: float64
```

Analysis: Constrained Mean–Variance Frontier and Tangency Portfolio ($r_f = 0$)

1) Differences in efficient frontier shape

In this constraint version of the problem, we introduced the inequality constraint $w \geq 0$. Since the optimization can't have negative weights, the feasible set shrinks relative to the unconstrained set. The position magnitudes are restricted as we can't finance long positions with short positions. It is also visible in the efficient frontier. The frontier is less extreme, which makes it more curved than the unconstrained version.

2) Sharpe ratio under constrained model

The orange star in the graph shows the tangency portfolio where the Sharpe ratio is highest among the feasible solutions for the constrained problem. The optimal Sharpe ratio that an investor could achieve without short selling is 0.44 for the local currency returns and nearly 0.45 for the USD-denominated returns. It is much lower than the unconstrained case since the no-short-selling constraint enables the investor's ability to exploit diversification using extreme short positions. In the new constrained model the efficient frontier model moves inward and the maximum Sharp ratio that can be achieved decreases.

3) Tangency portfolio under constraint model

The optimal portfolio in the new constrained model is consist of SPX and NKY. All other assets have zero weight.

- **Local-currency tangency:**
 $w_{\text{SPX}} \approx 0.886, \quad w_{\text{NKY}} \approx 0.114$
- **USD-denominated tangency:**
 $w_{\text{SPX}} \approx 0.822, \quad w_{\text{NKY}} \approx 0.178$

Since the problem can't go negatives in the weights it can only choose assets that will have the best combination of high expected return with the lowest volatility. The positions are bounded between 0 and 1. This create a more simple and stable portfolio with only 2 aseets rather than having extreme positions in each asset. The assets that have zero weights are dominated from the assets that have better risk-retun ratio.

4) Local-currency vs USD-denominated

Like the unconstrained model, changing the mean vector and covariance matrix makes the local currency frontier and tangency differ from the USD-denominated one. The 2 assets that create the optimal portfolio remains same; however, the weights change slightly (more weight towards NKY). This clearly indicates that currency denomination could affect the optimization model.

5) Key takeaways

The key point in adding a no-short-selling constraint in the problem eliminates extremely large positions and bounds the weights. As a result, the optimal portfolio becomes more concentrated and becomes more realistic. This restriction makes the model lose its flexibility, but it makes it more implementable in real life. This is particularly important since there are many investors such as pension funds, and mutual funds operates under no-short selling clause. Although theoretically unconstrained version perform more efficiently, the constrained version could be more relevant for some investors. One extension for this case could be limiting the allowed short position. This could result in having more flexibility while ensuring that we don't have extreme positions.

Q2 — Final Summary and Conclusions

This question applied the classical mean–variance framework to a set of international equity indices, with portfolio performance evaluated in USD-denominated returns, as required when constructing an investor-facing portfolio. The results clearly illustrate how portfolio optimality depends both on currency denomination and on portfolio constraints, and they highlight the practical limitations of unconstrained mean–variance optimization.

USD-denominated tangency portfolios: unconstrained vs constrained

Unconstrained (short-selling allowed)

In the unconstrained case, the USD-denominated tangency portfolio achieves a Sharpe ratio of approximately **1.14**, substantially higher than what is attainable under realistic constraints. However, this performance is driven by extreme and highly leveraged weights. For example, the unconstrained USD tangency portfolio assigns very large positive exposures to:

- **DAX** (~4.03)
- **CAC** (~3.08)
- **SPX** (~1.73)

These positions are financed through large short positions, most notably:

- **EuroStoxx50** (~-6.68)
- Additional shorts in **UKX** and **SPTSX**

Although the weights sum to one, the portfolio embeds significant gross leverage and offsetting positions, making it highly sensitive to estimation error and impractical to implement.

Constrained (no short-selling, ($w \geq 0$))

By contrast, imposing the no short-sale constraint produces a dramatically different outcome. The USD-denominated constrained tangency portfolio achieves a Sharpe ratio of roughly **0.45**, less than half of the unconstrained value. The optimal portfolio becomes highly concentrated, allocating approximately:

- **82% to SPX**
- **18% to NKY**

All other assets receive zero weight. This reflects the optimizer's inability to use short positions to engineer synthetic exposures: instead, it selects the small subset of assets that offer the best risk-return trade-off under non-negativity constraints.

Interpretation of the weight behavior

The stark contrast in weights across the two models is not a coding artifact, but a fundamental property of the optimization problem. In the unconstrained model, the optimizer amplifies small differences in expected returns and correlations by constructing large offsetting long-short positions, leading to extreme leverage and unstable portfolios. Once short-selling is prohibited, these extreme solutions disappear, and the optimizer settles on simple, corner solutions with a few dominant assets and much lower achievable Sharpe ratios.

Role of currency denomination

Comparing local-currency and USD-denominated results shows that FX exposure materially affects optimal portfolios. In both unconstrained and constrained settings, converting returns to USD alters the estimated mean vector and covariance matrix, shifting both the efficient frontier and the tangency portfolio. In the constrained case, while the identity of the dominant assets remains the same (SPX and NKY), the weights shift toward NKY in USD terms, reflecting the additional diversification or risk introduced by currency movements.

Taken together, Q2 demonstrates that:

1. Mean–variance optimization is highly sensitive to constraints: unconstrained solutions can look very attractive in-sample but rely on extreme leverage.
2. Realistic constraints fundamentally change portfolio composition, producing simpler, more stable, but lower-Sharpe portfolios.
3. Currency denomination matters for portfolio construction, not just for reporting returns.
4. The classical Markowitz framework is best viewed as a theoretical benchmark, whose outputs must be tempered with economic intuition and implementation constraints.

These results motivate the use of constrained optimization in practice and set the stage for further analysis, such as rolling-window estimation and comparisons with market-cap-weighted portfolios in subsequent questions.

Q3 Portfolio Comparison

This question implements a rolling 5-year mean–variance portfolio optimization using quadratic programming (`quadprog`).

- Returns are computed at daily frequency and annualized.
- Portfolio weights are re-estimated using a 5-year rolling window.
- A long-only constraint is imposed in the baseline specification.
- The optimization is solved in the `quadprog` framework.

Justification for RF Rates

Risk-Free Rates and Their Role in Portfolio Construction

The risk-free rate represents the theoretical return on an investment with zero risk, typically proxied by government treasury bills or similar short-term securities. In portfolio construction, the risk-free rate serves as the benchmark for calculating excess returns—the returns above what could be earned risk-free. Optimal portfolio weights depend critically on these excess returns, as they determine the risk premium offered by each asset.

USD Investor Perspective

For this analysis, we adopt the perspective of a **USD-based investor** evaluating international equity markets. Since all asset prices in our dataset are denominated in USD (after currency conversion), we use a single USD risk-free rate applied uniformly across all assets. This approach is standard in international portfolio analysis and reflects the opportunity cost faced by USD investors: they can either invest in risky international equities or hold US Treasury bills.

Key insight: When returns are measured in USD, the currency movements (including interest rate differentials between countries) are already embedded in those returns. For example, the USD return on Ibovespa includes both:

- The local (BRL) return of the Brazilian stock market
- The appreciation or depreciation of BRL vs USD

Therefore, comparing USD returns to a USD risk-free rate gives the appropriate measure of excess return for a USD investor.

Sensitivity Analysis Through Multiple USD Rate Assumptions

Since the precise risk-free rate varies over time and depends on maturity choice, we conduct a sensitivity analysis using six different USD risk-free rate assumptions: 0%, 1%, 2%, 3%, 4%, and 5% annually. This range captures various potential overnight to short-term Treasury rates and allows us to assess how sensitive our optimal portfolio allocations are to risk-free rate assumptions. By calculating excess returns as log returns minus the risk-free rate for each scenario, we can evaluate whether our portfolio construction results are robust across different rate environments.

Initial Analysis

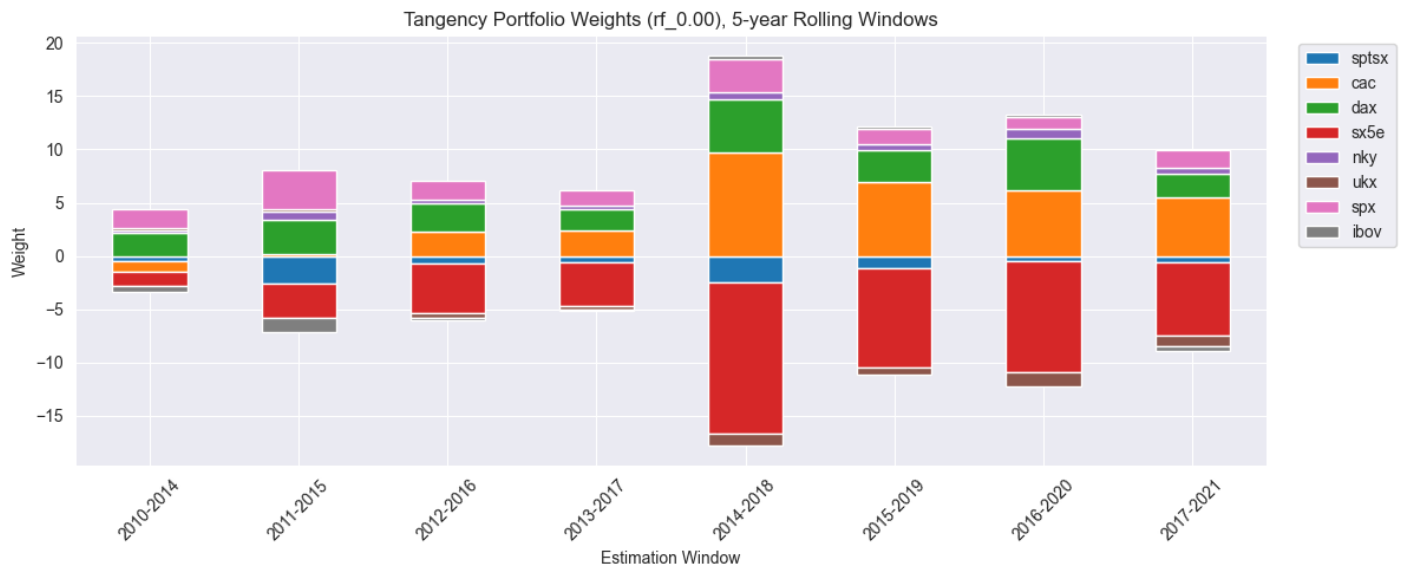
We begin by calculating the Risk Premium and Volatility across indices. The data we are working with has been converted into USD as explained aboved.

rf_0.00	0.0004	0.1714	0.0970	0.0941
rf_0.01	-0.0096	0.1614	0.0870	0.0841
rf_0.02	-0.0196	0.1514	0.0770	0.0742
rf_0.03	-0.0296	0.1414	0.0670	0.0642
rf_0.04	-0.0396	0.1314	0.0570	0.0542
rf_0.05	-0.0496	0.1214	0.0470	0.0442

Checking Constant Vol

rf_0.00	0.3410	0.2031	0.1826	0.2159
rf_0.01	0.3410	0.2031	0.1826	0.2159
rf_0.02	0.3410	0.2031	0.1826	0.2159
rf_0.03	0.3410	0.2031	0.1826	0.2159
rf_0.04	0.3410	0.2031	0.1826	0.2159
rf_0.05	0.3410	0.2031	0.1826	0.2159

Unconstrained Portfolio Construction Across RF Rates - 5 Year Rolling Window

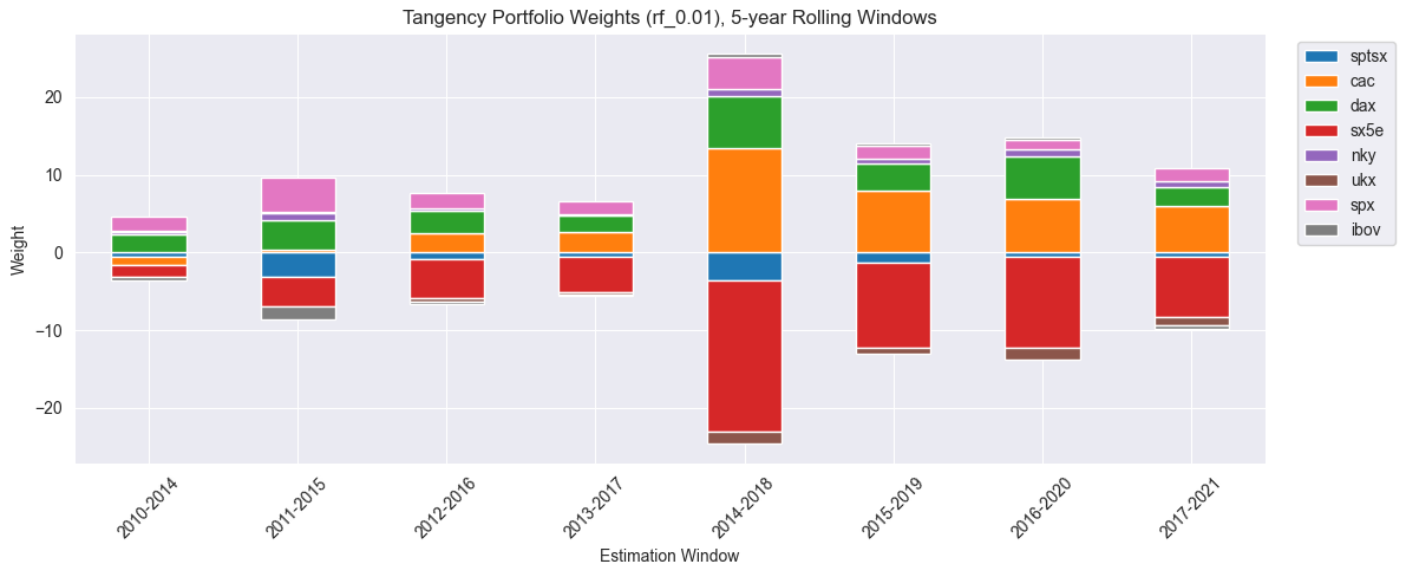


rf_0.00 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.44828	-1.03849	2.19567	-1.34568	0.20980	0.25781	1.70721	-0.53805
-2.60139	0.22385	3.18128	-3.17400	0.77054	0.21832	3.70703	-1.32562
-0.73992	2.33672	2.64878	-4.56926	0.30529	-0.52366	1.76821	-0.22615
-0.55635	2.43144	1.98733	-4.18228	0.25259	-0.31910	1.48807	-0.10170
-2.48918	9.73497	4.94385	-14.20121	0.64845	-1.11758	3.18027	0.30042
-1.10114	6.91041	2.98311	-9.41398	0.54587	-0.63219	1.45531	0.25262
-0.44006	6.16017	4.92046	-10.43916	0.83294	-1.39584	1.13937	0.22212
-0.54950	5.45338	2.21069	-6.92223	0.63289	-1.00603	1.59912	-0.41833

rf_0.00 (Annual rate: 0.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.567801	0.284517	1.995666
2011-2015	1.274469	0.618156	2.061725
2012-2016	0.538609	0.262333	2.053151
2013-2017	0.413885	0.198371	2.086420
2014-2018	0.825969	0.556335	1.484661
2015-2019	0.563450	0.325403	1.731547
2016-2020	0.612370	0.403127	1.519052
2017-2021	0.491149	0.319994	1.534870



rf_0.01 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.50491	-1.09438	2.38640	-1.46754	0.20176	0.24896	1.80925	-0.57955
-3.17863	0.32441	3.83496	-3.83093	0.87126	0.20563	4.35076	-1.57746
-0.83313	2.54933	2.88311	-4.96618	0.31633	-0.58345	1.87747	-0.24349
-0.62171	2.61105	2.12552	-4.47682	0.26001	-0.35088	1.56091	-0.10808
-3.54201	13.40325	6.76198	-19.47706	0.81201	-1.59872	4.21570	0.42484
-1.32900	8.02274	3.44043	-10.89052	0.59554	-0.75639	1.61786	0.29935
-0.50855	6.89889	5.51554	-11.68157	0.88162	-1.59211	1.22781	0.25838
-0.61787	6.01739	2.44971	-7.63848	0.64888	-1.13441	1.72957	-0.45479

rf_0.01 (Annual rate: 1.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.595995	0.303827	1.961627
2011-2015	1.507121	0.736316	2.046841
2012-2016	0.568773	0.282080	2.016354
2013-2017	0.427039	0.209606	2.037346
2014-2018	1.119118	0.761772	1.469098
2015-2019	0.633545	0.372076	1.702728
2016-2020	0.664707	0.444503	1.495392
2017-2021	0.520348	0.345798	1.504774

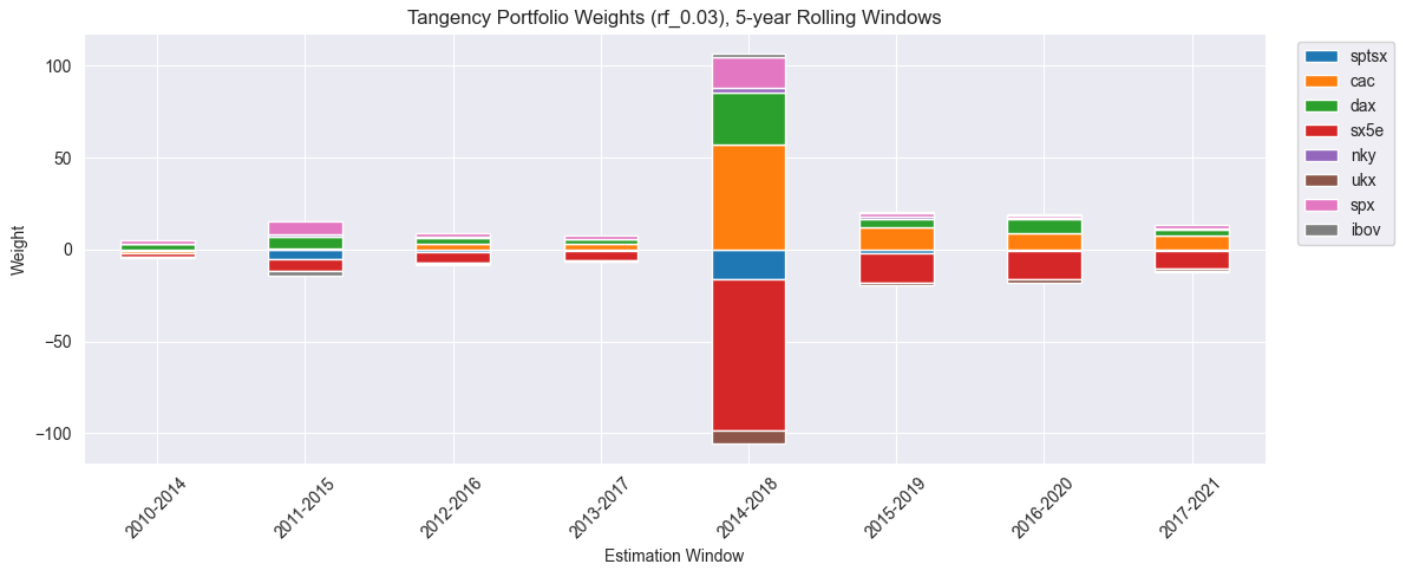


rf_0.02 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.57224	-1.16084	2.61320	-1.61244	0.19221	0.23842	1.93058	-0.62889
-4.04410	0.47518	4.81504	-4.81589	1.02228	0.18660	5.31593	-1.95504
-0.94588	2.80651	3.16656	-5.44630	0.32970	-0.65577	2.00964	-0.26446
-0.69876	2.82278	2.28842	-4.82402	0.26876	-0.38835	1.64677	-0.11560
-5.90606	21.64011	10.84446	-31.32360	1.17926	-2.67908	6.54069	0.70422
-1.64546	9.56756	4.07556	-12.94116	0.66453	-0.92888	1.84362	0.36425
-0.59574	7.83919	6.27301	-13.26303	0.94359	-1.84193	1.34038	0.30452
-0.70180	6.70984	2.74316	-8.51785	0.66850	-1.29202	1.88973	-0.49955

rf_0.02 (Annual rate: 2.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.631412	0.327177	1.929880
2011-2015	1.860938	0.914654	2.034581
2012-2016	0.607357	0.306390	1.982300
2013-2017	0.444333	0.223163	1.991072
2014-2018	1.789817	1.227204	1.458451
2015-2019	0.734782	0.437928	1.677859
2016-2020	0.734055	0.497970	1.474095
2017-2021	0.558474	0.378093	1.477081



rf_0.03 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.65364	-1.24118	2.88735	-1.78760	0.18066	0.22569	2.07724	-0.68853
-5.48551	0.72628	6.44733	-6.45632	1.27380	0.15491	6.92340	-2.58389
-1.08503	3.12391	3.51640	-6.03885	0.34619	-0.74503	2.17276	-0.29034
-0.79093	3.07607	2.48331	-5.23940	0.27922	-0.43317	1.74950	-0.12460
-16.07382	57.06684	28.40320	-82.27558	2.75880	-7.32573	16.54045	1.90583
-2.11466	11.85797	5.01723	-15.98152	0.76681	-1.18462	2.17833	0.46047
-0.71048	9.07664	7.26985	-15.34425	1.02514	-2.17070	1.48853	0.36526
-0.80731	7.58025	3.11203	-9.62323	0.69317	-1.49014	2.09105	-0.55582

rf_0.03 (Annual rate: 3.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.676314	0.355854	1.900541
2011-2015	2.456866	1.213273	2.024991
2012-2016	0.657319	0.336891	1.951131
2013-2017	0.466987	0.239751	1.947799
2014-2018	4.707502	3.240230	1.452830
2015-2019	0.889707	0.536900	1.657118
2016-2020	0.828478	0.569297	1.455266
2017-2021	0.608968	0.419421	1.451927



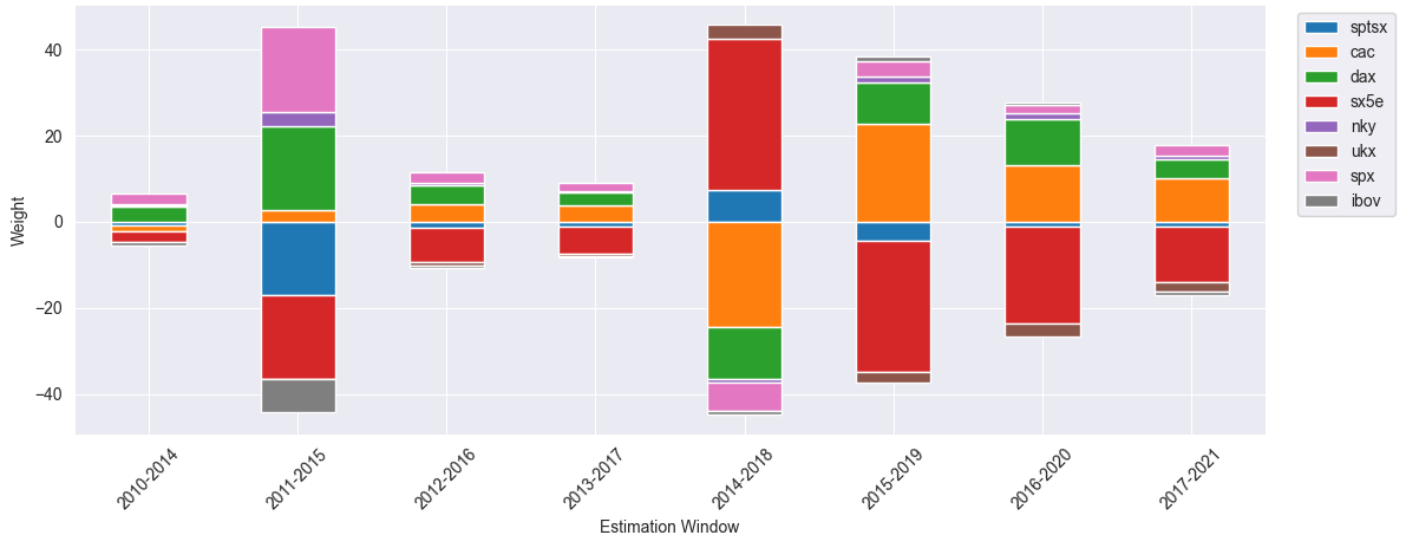
rf_0.04 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.75401	-1.34025	3.22544	-2.00361	0.16642	0.21000	2.25811	-0.76209
-8.36448	1.22782	9.70755	-9.73278	1.77616	0.09161	10.13402	-3.83990
-1.26109	3.52551	3.95903	-6.78858	0.36705	-0.85796	2.37914	-0.32309
-0.90318	3.38453	2.72063	-5.74522	0.29196	-0.48776	1.87459	-0.13556
25.35530	-87.28138	-43.14088	125.33112	-3.67714	11.60729	-24.20414	-2.99018
-2.88233	15.60545	6.55795	-20.95604	0.93416	-1.60305	2.72597	0.61790
-0.86828	10.77852	8.64082	-18.20656	1.13730	-2.62287	1.69229	0.44878
-0.94393	8.70735	3.58967	-11.05457	0.72511	-1.74669	2.35174	-0.62868

rf_0.04 (Annual rate: 4.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.734018	0.391744	1.873721
2011-2015	3.657099	1.812141	2.018110
2012-2016	0.723186	0.376074	1.922986
2013-2017	0.496750	0.260388	1.907731
2014-2018	-7.231511	4.979384	-1.452290
2015-2019	1.149551	0.700664	1.640661
2016-2020	0.962091	0.668584	1.438999
2017-2021	0.677303	0.473822	1.429445

Tangency Portfolio Weights (rf_0.05), 5-year Rolling Windows

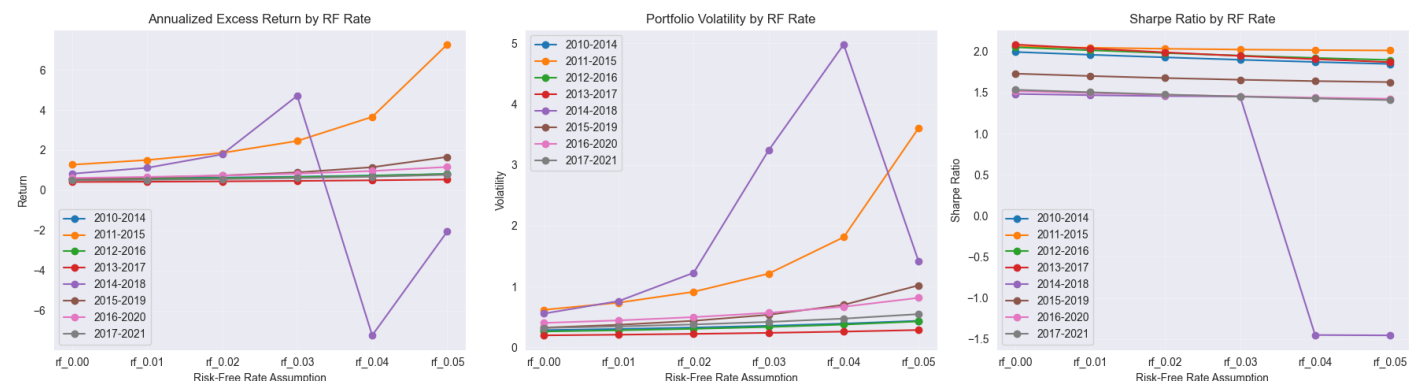


rf_0.05 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.88089	-1.46547	3.65276	-2.27664	0.14842	0.19015	2.48672	-0.85506
-16.97063	2.72707	19.45337	-19.52715	3.27789	-0.09761	19.73160	-7.59454
-1.49099	4.04991	4.53701	-7.76758	0.39430	-1.00543	2.64864	-0.36586
-1.04284	3.76834	3.01594	-6.37462	0.30782	-0.55568	2.03025	-0.14920
7.35047	-24.54857	-12.04829	35.10660	-0.88012	3.37912	-6.49681	-0.86241
-4.36618	22.84894	9.53602	-30.57127	1.25762	-2.41184	3.78450	0.92220
-1.09898	13.26668	10.64518	-22.39128	1.30127	-3.28393	1.99017	0.57089
-1.12781	10.22431	4.23254	-12.98103	0.76810	-2.09198	2.70261	-0.72674

rf_0.05 (Annual rate: 5.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.809594	0.437729	1.849531
2011-2015	7.264865	3.607248	2.013964
2012-2016	0.812252	0.427951	1.898002
2013-2017	0.536230	0.286590	1.871072
2014-2018	-2.057237	1.412124	-1.456838
2015-2019	1.661131	1.019964	1.628616
2016-2020	1.162055	0.815258	1.425384
2017-2021	0.772734	0.548130	1.409765



Average Metrics Across All Windows:

	Return	Volatility	Sharpe
RF_Rate			
rf_0.00	0.66096	0.37103	1.80839
rf_0.01	0.75458	0.43200	1.77927
rf_0.02	0.92015	0.53907	1.75316
rf_0.03	1.41152	0.86395	1.73020
rf_0.04	0.14606	1.20785	1.34742
rf_0.05	1.37020	1.06937	1.32994

Analysis of Unconstrained Portfolio (Short-Selling Allowed)

The unconstrained mean-variance optimization produces portfolios with extreme long and short positions, demonstrating the well-known instability of unconstrained Markowitz optimization in practice.

Weight Characteristics and Leverage

Examining the tangency portfolio weights across different risk-free rate scenarios reveals several striking patterns:

Extreme Positions: The unconstrained portfolios exhibit substantial leverage through large long and short positions. For example, in the 2014-2018 window at $rf=0.00$, we observe:

- Long positions: DAX (9.73), CAC (9.73), SPX (3.18)
- Short positions: SX5E (-14.20), TSX (-2.49), UKX (-1.12)

These weights sum to 1.0 (satisfying the budget constraint) but involve gross exposure far exceeding 100%. The sum of absolute weights in this example exceeds 500%, indicating a highly leveraged portfolio requiring substantial margin and borrowing.

Asset Selection Patterns: Certain patterns emerge consistently across estimation windows:

- European indices show extreme offsetting positions: DAX and CAC receive large positive weights while SX5E (EuroStoxx 50) receives large negative weights. This reflects the optimization exploiting correlation structure within European markets creating a synthetic “spread” position that aims to profit from relative performance differences between highly correlated indices.
- SPX consistently receives positive weight: The S&P 500 appears with positive exposure across all windows, reflecting its favorable risk-return profile in the estimation samples.
- Emerging markets (IBOV) show mixed positions: Positions in the Brazilian market vary from small positive to small negative, suggesting its risk-return characteristics are less compelling relative to developed markets.

Sensitivity to Risk-Free Rate Assumptions

As the risk-free rate increases from 0% to 5%, the magnitude of leverage increases dramatically:

- At $rf=0.00$: Portfolio weights range from approximately -15 to +10
- At $rf=0.05$: Portfolio weights range from approximately -30 to +22

This amplification occurs because higher risk-free rates reduce the excess returns (risk premiums) of all assets. To achieve the maximum Sharpe ratio, the optimizer compensates by increasing leverage—taking larger long and short positions to exploit the remaining (smaller) differences in risk-adjusted returns.

Performance Metrics

The unconstrained portfolios deliver impressive in-sample performance: Average Sharpe Ratios (across all windows):

- $r_f=0.00$: 1.808
- $r_f=0.03$: 1.730
- $r_f=0.05$: 1.330

These Sharpe ratios substantially exceed those of the constrained portfolios, but this comes at the cost of:

- Extreme leverage requiring substantial capital and margin
- High sensitivity to estimation error in expected returns
- Significant transaction costs from large positions and frequent rebalancing
- Practical infeasibility in most real-world investment contexts

Temporal Instability

Portfolio weights exhibit high variability across estimation windows. For instance, examining SPX weights at $r_f=0.00$:

- 2010-2014: 1.71
- 2014-2018: 3.18
- 2017-2021: 1.60

This instability reflects the sensitivity of mean-variance optimization to changes in the estimated mean vector and covariance matrix. Small changes in these inputs can lead to dramatic shifts in optimal weights, particularly in the unconstrained setting where there are no bounds preventing extreme allocations.

Economic Interpretation

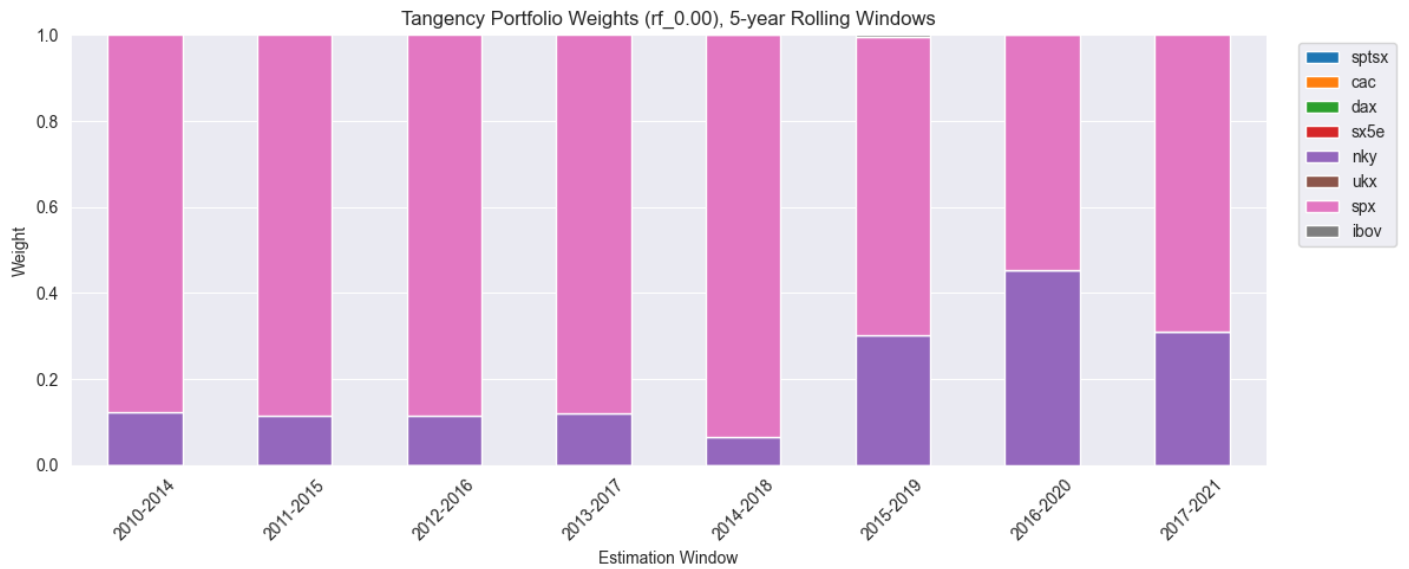
The extreme positions reflect the optimizer's attempt to:

- Exploit correlation structure: Creating synthetic exposures through combinations of correlated assets
- Maximize diversification benefits: Using short positions to hedge specific risk factors
- Amplify small differences in Sharpe ratios: Through leverage, exploiting even minor advantages in risk-adjusted returns

However, these positions rest on several strong assumptions:

- Stable correlation structure: Relationships between markets remain constant out-of-sample
- Accurate return estimates: The sample means are unbiased estimates of true expected returns (highly questionable over 5-year windows)
- No transaction costs: Trading and rebalancing is costless
- No market impact: Large positions can be established without moving prices
- Unlimited leverage: Borrowing and short-selling at the risk-free rate is possible

Constrained Portfolio (Long-Only) Optimisation Across RF Rates



rf_0.00 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
0.0	-0.0	0.0	0.0	0.12337	0.0	0.87663	0.00000
0.0	0.0	0.0	-0.0	0.11559	0.0	0.88441	0.00000
-0.0	0.0	0.0	-0.0	0.11411	-0.0	0.88589	-0.00000
0.0	0.0	0.0	-0.0	0.12008	0.0	0.87992	-0.00000
-0.0	0.0	-0.0	-0.0	0.06577	-0.0	0.93423	0.00000
0.0	-0.0	-0.0	-0.0	0.30249	-0.0	0.69320	0.00431
-0.0	0.0	0.0	-0.0	0.45460	0.0	0.54522	0.00018
0.0	-0.0	-0.0	0.0	0.31165	0.0	0.68835	-0.00000

rf_0.00 (Annual rate: 0.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.130864	0.156762	0.834794
2011-2015	0.106784	0.153260	0.696749
2012-2016	0.128142	0.129697	0.988013
2013-2017	0.143008	0.119193	1.199804
2014-2018	0.065571	0.131287	0.499448
2015-2019	0.096653	0.122494	0.789042
2016-2020	0.126920	0.162752	0.779838
2017-2021	0.148206	0.165612	0.894899

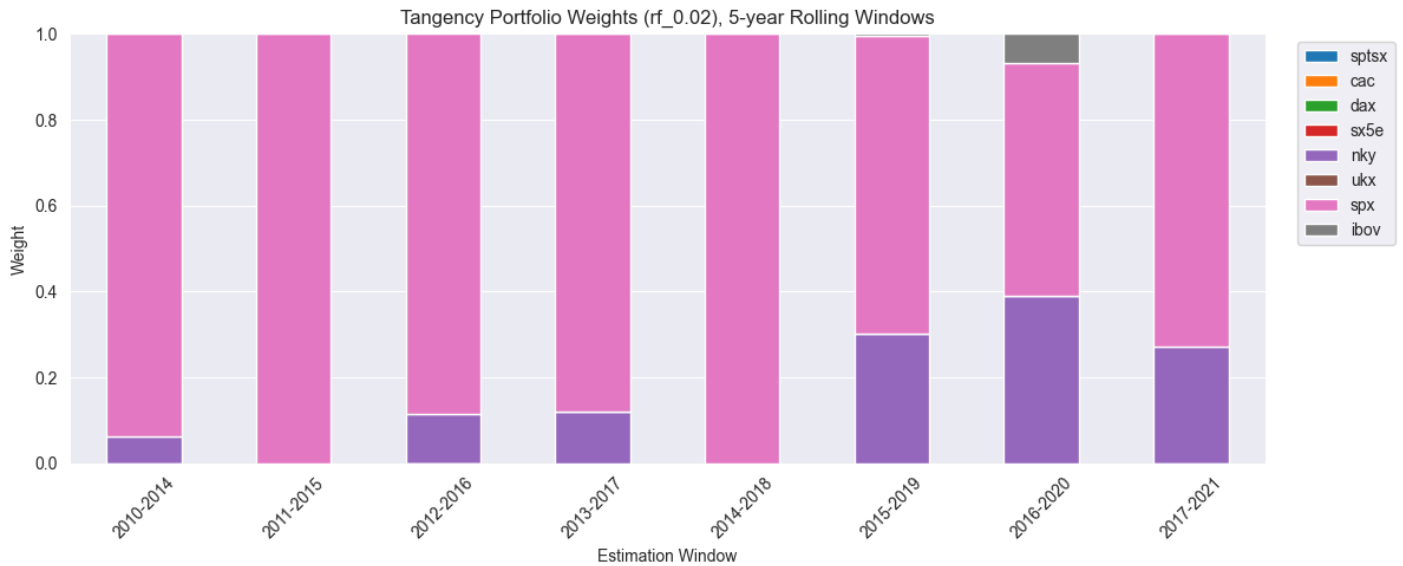


rf_0.01 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
-0.0	0.0	-0.0	-0.0	0.06169	0.0	0.93831	0.00000
0.0	0.0	-0.0	-0.0	0.11559	0.0	0.88441	0.00000
-0.0	-0.0	0.0	-0.0	0.11411	0.0	0.88589	0.00000
0.0	-0.0	0.0	-0.0	0.12008	0.0	0.87992	0.00000
0.0	0.0	-0.0	0.0	0.06577	-0.0	0.93423	0.00000
-0.0	0.0	0.0	-0.0	0.30249	-0.0	0.69320	0.00431
0.0	0.0	0.0	-0.0	0.45460	0.0	0.54522	0.00018
-0.0	0.0	-0.0	0.0	0.31165	-0.0	0.68835	-0.00000

rf_0.01 (Annual rate: 1.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.126219	0.163468	0.772132
2011-2015	0.096784	0.153260	0.631502
2012-2016	0.118142	0.129697	0.910911
2013-2017	0.133008	0.119193	1.115908
2014-2018	0.055571	0.131287	0.423280
2015-2019	0.086653	0.122494	0.707407
2016-2020	0.116920	0.162752	0.718396
2017-2021	0.138206	0.165612	0.834518

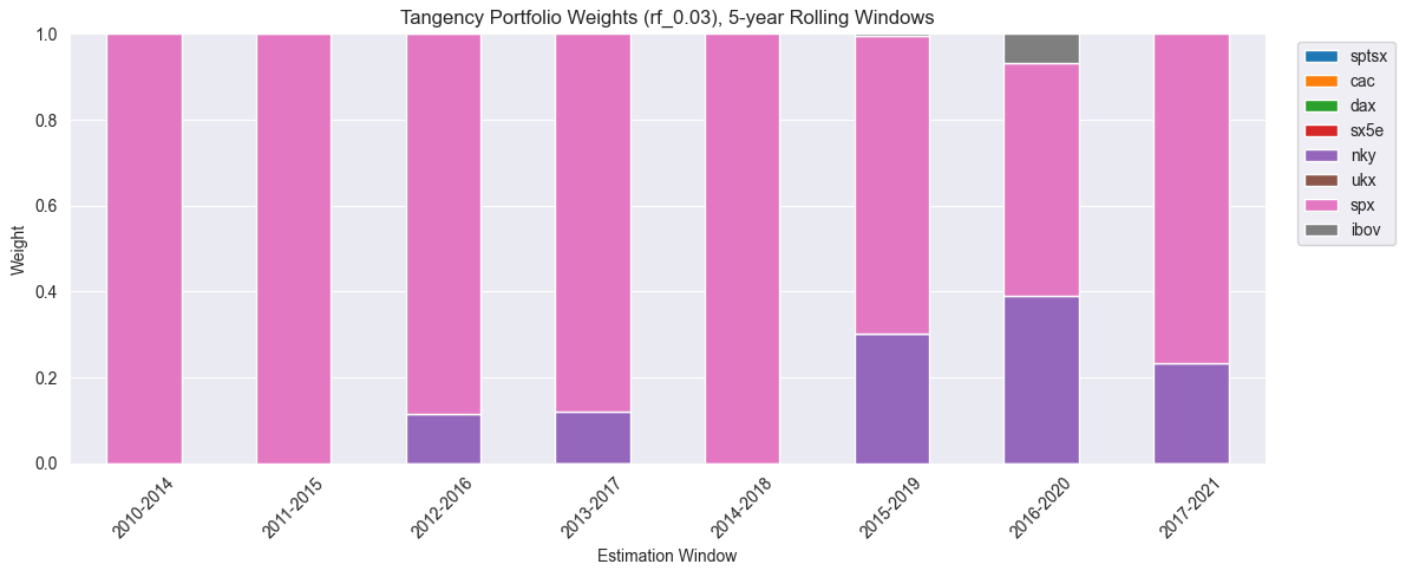


rf_0.02 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
0.0	-0.0	0.0	0.0	0.06169	0.0	0.93831	0.00000
0.0	0.0	0.0	-0.0	0.00002	0.0	0.99998	0.00000
0.0	0.0	-0.0	0.0	0.11411	0.0	0.88589	-0.00000
-0.0	0.0	0.0	-0.0	0.12008	0.0	0.87992	0.00000
0.0	0.0	0.0	-0.0	0.00003	-0.0	0.99997	0.00000
-0.0	-0.0	-0.0	-0.0	0.30249	0.0	0.69320	0.00431
0.0	-0.0	0.0	-0.0	0.38964	0.0	0.54362	0.06673
0.0	0.0	0.0	0.0	0.27269	0.0	0.72731	0.00000

rf_0.02 (Annual rate: 2.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.116220	0.163468	0.710961
2011-2015	0.093792	0.165242	0.567608
2012-2016	0.108143	0.129697	0.833813
2013-2017	0.123009	0.119193	1.032016
2014-2018	0.047732	0.136801	0.348913
2015-2019	0.076654	0.122494	0.625776
2016-2020	0.109861	0.167195	0.657083
2017-2021	0.131105	0.169092	0.775347

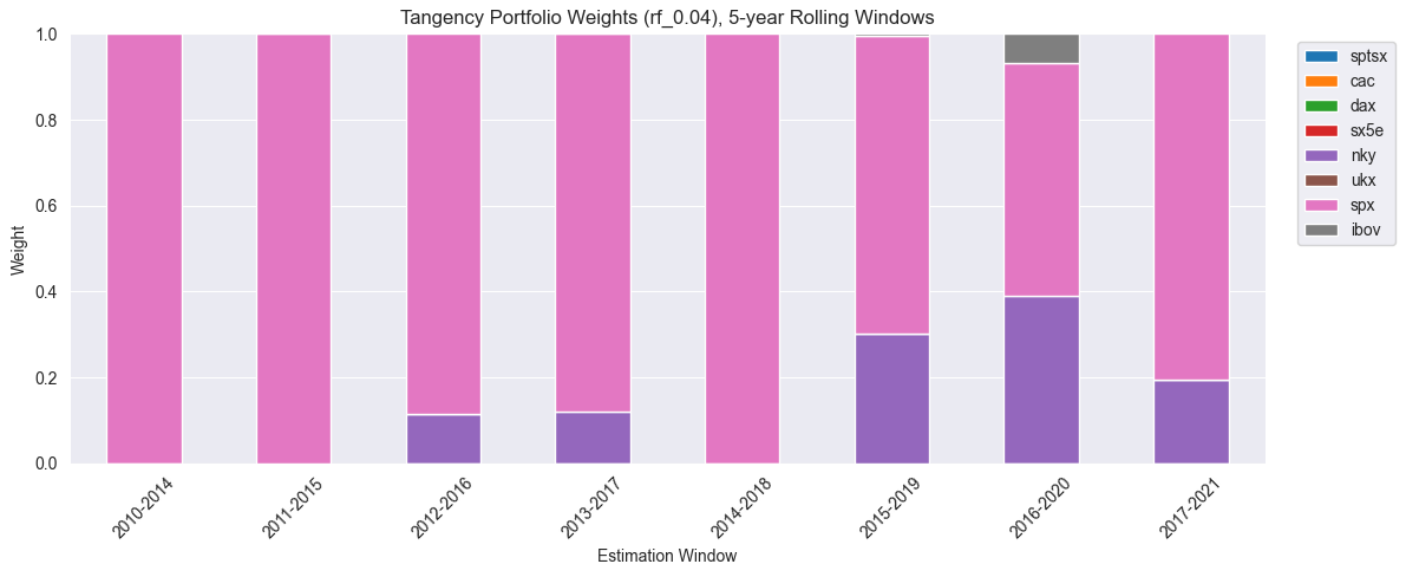


rf_0.03 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
0.0	0.0	0.0	-0.0	0.00001	0.0	0.99999	0.00000
0.0	0.0	-0.0	-0.0	0.00002	0.0	0.99998	0.00000
0.0	-0.0	-0.0	0.0	0.11411	0.0	0.88589	0.00000
0.0	-0.0	0.0	0.0	0.12008	-0.0	0.87992	-0.00000
0.0	0.0	-0.0	-0.0	0.00003	-0.0	0.99997	0.00000
-0.0	0.0	0.0	-0.0	0.30249	0.0	0.69320	0.00431
-0.0	0.0	0.0	-0.0	0.38964	-0.0	0.54362	0.06673
0.0	0.0	0.0	-0.0	0.23374	0.0	0.76626	-0.00000

rf_0.03 (Annual rate: 3.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.111576	0.171394	0.650990
2011-2015	0.083793	0.165242	0.507097
2012-2016	0.098144	0.129697	0.756717
2013-2017	0.113010	0.119193	0.948126
2014-2018	0.037732	0.136801	0.275821
2015-2019	0.066655	0.122494	0.544147
2016-2020	0.099862	0.167195	0.597278
2017-2021	0.124004	0.172966	0.716925

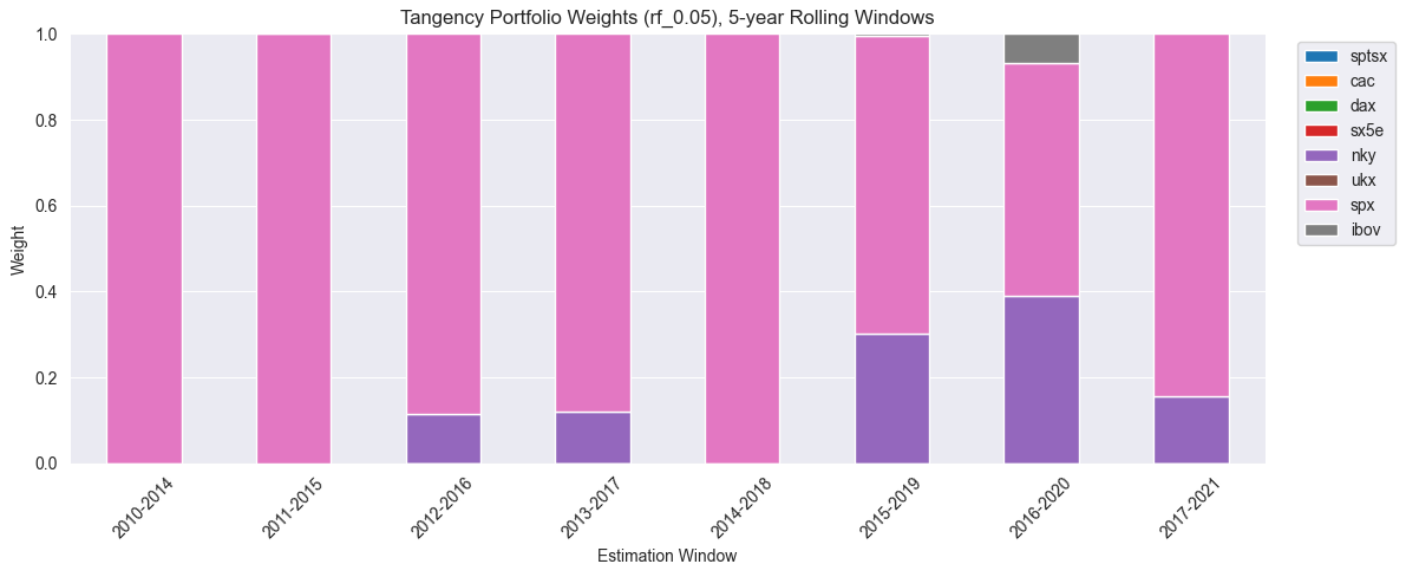


rf_0.04 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
0.0	0.0	-0.0	0.0	0.00001	0.0	0.99999	-0.00000
0.0	-0.0	-0.0	0.0	0.00002	0.0	0.99998	0.00000
-0.0	0.0	-0.0	0.0	0.11411	0.0	0.88589	-0.00000
-0.0	0.0	0.0	0.0	0.12008	-0.0	0.87992	0.00000
0.0	0.0	0.0	0.0	0.00003	-0.0	0.99997	0.00000
-0.0	0.0	0.0	0.0	0.30249	0.0	0.69320	0.00431
-0.0	0.0	0.0	-0.0	0.38964	-0.0	0.54362	0.06673
0.0	0.0	0.0	-0.0	0.19479	0.0	0.80521	-0.00000

rf_0.04 (Annual rate: 4.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.101577	0.171394	0.592653
2011-2015	0.073795	0.165242	0.446588
2012-2016	0.088145	0.129697	0.679625
2013-2017	0.103011	0.119193	0.864240
2014-2018	0.027734	0.136801	0.202732
2015-2019	0.056656	0.122494	0.462522
2016-2020	0.089863	0.167195	0.537476
2017-2021	0.116903	0.177209	0.659689

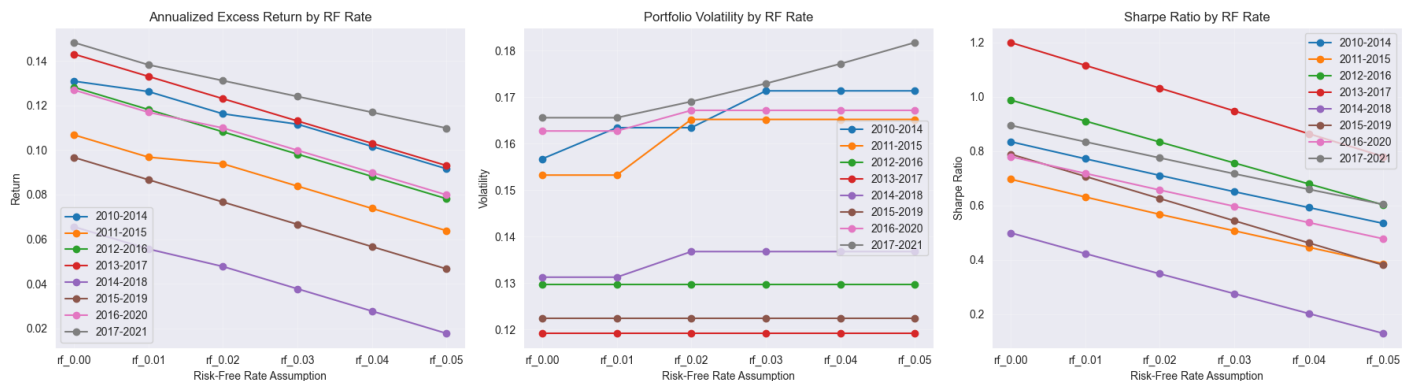


rf_0.05 - Detailed Weights:

sptsx	cac	dax	sx5e	nky	ukx	spx	ibov
0.0	0.0	-0.0	0.0	0.00001	0.0	0.99999	-0.00000
-0.0	0.0	0.0	0.0	0.00002	0.0	0.99998	0.00000
0.0	-0.0	0.0	0.0	0.11411	-0.0	0.88589	0.00000
0.0	0.0	0.0	0.0	0.12008	-0.0	0.87992	0.00000
-0.0	0.0	0.0	-0.0	0.00003	0.0	0.99997	0.00000
0.0	0.0	-0.0	0.0	0.30249	0.0	0.69320	0.00431
0.0	-0.0	-0.0	0.0	0.38964	-0.0	0.54362	0.06673
-0.0	-0.0	-0.0	-0.0	0.15583	-0.0	0.84417	-0.00000

rf_0.05 (Annual rate: 5.0%) - Portfolio Metrics:

Window	Return	Volatility	Sharpe
2010-2014	0.091579	0.171394	0.534319
2011-2015	0.063797	0.165242	0.386081
2012-2016	0.078147	0.129697	0.602536
2013-2017	0.093013	0.119193	0.780358
2014-2018	0.017736	0.136801	0.129646
2015-2019	0.046658	0.122494	0.380900
2016-2020	0.079865	0.167195	0.477676
2017-2021	0.109803	0.181796	0.603990



Average Metrics Across All Windows:

	Return	Volatility	Sharpe
RF_Rate			
rf_0.00	0.11827	0.14263	0.83532
rf_0.01	0.10894	0.14347	0.76426
rf_0.02	0.10081	0.14665	0.69394
rf_0.03	0.09185	0.14812	0.62464
rf_0.04	0.08221	0.14865	0.55569
rf_0.05	0.07257	0.14923	0.48694

Constrained Portfolio Analysis (Long-Only)

The long-only constraint fundamentally transforms the optimization outcome, producing portfolios that are practical, implementable, and far more stable—though at the cost of lower in-sample Sharpe ratios.

Weight Characteristics and Asset Selection

The constrained portfolios exhibit dramatically different characteristics from their unconstrained counterparts:

Concentration in Two Assets: Across nearly all estimation windows and risk-free rate scenarios, the optimal long-only portfolio concentrates in just two indices:

- SPX (S&P 500): Typically 55-95% of portfolio weight
- NKY (Nikkei 225): Typically 5-45% of portfolio weight

For example, at $rf=0.00$ in the 2010-2014 window:

- SPX: 87.7%
- NKY: 12.3%
- All other assets: ~0%

This concentration pattern emerges because, within the long-only constraint, these two indices offer the most attractive risk-return combinations. The optimizer selects them through a process of elimination:

- SPX dominance: The S&P 500 delivers the highest Sharpe ratio among developed market indices in most estimation windows, explaining its consistent large allocation.
- NKY as diversifier: The Nikkei 225 has relatively lower correlation with SPX compared to European indices (correlation 0.10-0.25 vs. 0.60+ for European markets), providing meaningful diversification benefits.
- Other assets excluded: European indices (CAC, DAX, SX5E, UKX), TSX, and IBOV are effectively crowded out—they either have inferior standalone Sharpe ratios or are too highly correlated with SPX to provide additional diversification value.

Temporal Variation

While far more stable than unconstrained portfolios, the long-only weights still show meaningful variation across estimation windows:

SPX weight variation (at $r_f=0.00$):

- 2010-2014: 87.7%
- 2014-2018: 93.4%
- 2016-2020: 54.5%
- 2017-2021: 68.8%

The 2016-2020 window is particularly notable, showing the highest allocation to NKY (45.5%). This period includes the COVID-19 pandemic onset and may reflect a period where Japanese equities offered more attractive risk-adjusted returns or greater diversification benefits.

Sensitivity to Risk-Free Rate

The long-only constraint creates an interesting dynamic as the risk-free rate increases:

Asset allocation patterns:

- Low RF rates (0-1%): Portfolios maintain meaningful diversification between SPX and NKY
- Higher RF rates (3-5%): Portfolios increasingly concentrate in SPX, with some windows showing near 100% allocation to SPX

This occurs because higher risk-free rates effectively raise the “hurdle rate” for inclusion. Assets with marginal risk-adjusted returns fall below this hurdle and are eliminated from the portfolio. Since SPX typically has the highest Sharpe ratio, it survives this selection process while other assets are excluded.

Performance Metrics

The constrained portfolios show more modest but realistic performance: Average Sharpe Ratios (across all windows):

- $r_f=0.00$: 0.835
- $r_f=0.03$: 0.625
- $r_f=0.05$: 0.487

These Sharpe ratios are:

- 53-60% lower than unconstrained portfolios
- Still substantially higher than the market-cap benchmark (detailed in Section 3.4)
- More realistic and achievable in practical implementation

Return and Volatility Characteristics:

- Average annualized return: 9.6-11.8% (depending on RF rate)
- Average annualized volatility: 14.3-14.9%
- Both metrics remain relatively stable across different risk-free rate assumptions

Economic Rationale

The dominance of SPX and NKY in the long-only portfolios can be explained through several economic lenses:

- Home bias and market efficiency: The S&P 500 represents the largest, most liquid, and arguably most informationally efficient equity market globally. Its strong performance may reflect genuine economic fundamentals rather than estimation error.
- Diversification limits: The high correlations among developed markets (European indices correlate 0.80-0.95 with each other, and 0.60-0.70 with SPX) severely limit diversification benefits within a long-only framework. Adding European indices to an SPX-heavy portfolio provides minimal risk reduction.
- Currency effects: Since all returns are USD-denominated, the optimization implicitly accounts for currency risk. European indices share common EUR exposure, while NKY provides JPY diversification—this may partially explain NKY's consistent inclusion.
- Emerging market concerns: IBOV's higher standalone volatility (39% annualized) and only moderate correlation with other indices (0.54-0.69) is insufficient to justify inclusion given its lower average returns during the sample period.

Practical Advantages

The long-only portfolios offer several practical benefits:

- No leverage or margin requirements: Maximum loss is limited to invested capital
- Implementable with standard instruments: Can be executed with ETFs or index futures
- Lower transaction costs: Fewer assets and more stable weights reduce rebalancing costs
- Regulatory compliance: Suitable for investors with restrictions on short-selling
- Transparent risk: Easier to explain and monitor compared to complex long-short portfolios

Comparison: Unconstrained vs. Constrained Optimum Portfolios

The contrast between unconstrained and long-only portfolios illustrates fundamental trade-offs in portfolio optimization between theoretical optimality and practical feasibility.

Weight Structure Comparison

Unconstrained portfolios:

- Number of assets with meaningful positions: 6-8 assets (both long and short)
- Gross exposure: 300-800% of capital (depending on RF rate and window)
- Largest long position: Often 10-20x the largest short position in absolute value
- Herfindahl index of absolute weights: Would be extremely high, reflecting concentration despite multi-asset exposure

Constrained portfolios:

- Number of assets with meaningful positions: Typically 2 assets (SPX + NKY)
- Gross exposure: Exactly 100% of capital (fully invested, no leverage)
- Concentration: Herfindahl index 0.76
- Simplicity: Easier to implement and monitor

The constrained portfolios are paradoxically more concentrated (in terms of number of assets held long) but far less risky in terms of leverage and complexity.

Performance Comparison

The unconstrained portfolios deliver Sharpe ratios approximately 100-120% higher than constrained portfolios. However, this comparison is misleading for several reasons:

- Leverage normalization: The unconstrained Sharpe ratios are calculated on highly leveraged portfolios. If we were to compare on an equal-leverage basis (scaling the constrained portfolio to match the gross exposure of the unconstrained portfolio), the Sharpe ratio difference would narrow substantially.
- Estimation error sensitivity: The unconstrained performance is highly sensitive to small errors in expected return estimates. Mean-variance optimization is known to amplify estimation errors, particularly in the unconstrained case where the optimizer can take arbitrarily large positions based on small perceived advantages in expected returns.
- Out-of-sample degradation: Academic research consistently shows that unconstrained mean-variance portfolios suffer severe out-of-sample performance degradation. The in-sample Sharpe ratio advantage typically disappears or reverses out-of-sample.

Stability Comparison

Examining portfolio turnover and weight stability across rolling windows:

Unconstrained portfolios:

- Weight changes of 5-10 (in absolute terms) for individual assets across adjacent windows
- Complete sign reversals (long to short or vice versa) are common
- High implied transaction costs from constant rebalancing

Constrained portfolios:

- Weight changes of 0.05-0.30 for the two main assets across adjacent windows
- No sign reversals (impossible given long-only constraint)
- Implied lower transaction costs

The weight stability ratio (average absolute weight change) is approximately 20-30x higher for unconstrained versus constrained portfolios.

Risk Comparison

Unconstrained risks:

- Leverage risk: Potential for losses exceeding invested capital
- Liquidity risk: Large positions may be difficult to unwind in stress periods
- Model risk: Extreme sensitivity to optimization inputs
- Counterparty risk: Dependence on margin lending and short-selling arrangements
- Regulatory risk: Potential restriction on leverage or short-selling

Constrained risks:

- Concentration risk: Heavy reliance on two markets
- No downside protection: Long-only means no hedges against market declines
- Tracking error vs. market-cap benchmark: High active risk relative to traditional benchmarks

Unconstrained Portfolios - Impracticality

Despite their superior in-sample Sharpe ratios, unconstrained portfolios face severe practical limitations:

- Capital requirements: A portfolio with 500% gross exposure requires 5x the capital (including margin) compared to a long-only portfolio, dramatically reducing capital efficiency when accounting for margin costs.
- Estimation error: Academic research (DeMiguel, Garlappi, Uppal, 2009) shows that the theoretical gains from unconstrained optimization are typically consumed by estimation error. The constraint actually improves out-of-sample performance by limiting the optimizer's ability to exploit spurious patterns in historical data.
- Borrowing costs: Short positions and leverage involve borrowing costs that exceed the risk-free rate in practice. These costs are not reflected in the optimization but would significantly erode realized returns.
- Institutional constraints: Most institutional investors face regulatory or mandate restrictions on leverage and short-selling, making unconstrained portfolios simply infeasible regardless of their theoretical properties.

The Value of Constraints

The comparison illustrates a counter-intuitive principle in portfolio optimization: constraints can improve outcomes. By limiting the optimizer's freedom, we:

- Reduce estimation error amplification: Cannot take extreme positions based on small differences in estimated returns
- Improve out-of-sample robustness: Constrained portfolios tend to generalize better
- Enhance implementability: Simpler portfolios with lower transaction costs
- Increase stability: More predictable behavior across different market conditions

This finding aligns with extensive academic research showing that "naïve" diversification strategies (like equal-weighting) often outperform theoretically optimal unconstrained portfolios out-of-sample, precisely because they avoid the pitfalls of estimation error and over-optimization.

Market-Cap Portfolio Comparison (Given Weights)

We now compare the performance of the long-only portfolio to a market-cap based portfolio construction.

```
0 SPTSX INDEX CAC INDEX DAX INDEX SX5E INDEX NKY INDEX UKX INDEX SPX INDEX  \
1 0.041834 0.04318 0.036197 0.090955 0.092523 0.081589 0.602344

0 IBOV INDEX
1 0.011378
```

Return, Sharpe Ratio and other Performance Metrics

	Return			Volatility			\
	mean	std	min	max	mean	std	
RF_Rate							
rf_0.00	0.089239	0.031153	0.031219	0.134005	0.148353	0.023083	
rf_0.01	0.079240	0.031153	0.021219	0.124005	0.148353	0.023083	
rf_0.02	0.069240	0.031153	0.011220	0.114006	0.148353	0.023083	
rf_0.03	0.059241	0.031153	0.001221	0.104007	0.148353	0.023083	
rf_0.04	0.049243	0.031153	-0.008778	0.094008	0.148353	0.023083	
rf_0.05	0.039244	0.031153	-0.018776	0.084010	0.148353	0.023083	

	Sharpe					
	min	max	mean	std	min	max
RF_Rate						
rf_0.00	0.122796	0.1788	0.600719	0.200905	0.246880	0.898460
rf_0.01	0.122796	0.1788	0.531880	0.201217	0.167801	0.817026
rf_0.02	0.122796	0.1788	0.463044	0.202082	0.088725	0.735595
rf_0.03	0.122796	0.1788	0.394211	0.203492	0.009653	0.654167
rf_0.04	0.122796	0.1788	0.325380	0.205436	-0.069417	0.572743
rf_0.05	0.122796	0.1788	0.256552	0.207899	-0.148483	0.491321

	MAD	RMSE	Tang_Herf	MktCap_Herf
RF_Rate				
rf_0.00	0.078764	0.112903	0.711221	0.391363
rf_0.01	0.079728	0.115364	0.723787	0.391363
rf_0.02	0.084373	0.120580	0.762227	0.391363
rf_0.03	0.086300	0.123400	0.781501	0.391363
rf_0.04	0.086300	0.123803	0.787067	0.391363
rf_0.05	0.086300	0.124669	0.793391	0.391363

Analysis of the Market-Cap Portfolio Performance

The market-cap weighted portfolio serves as a natural benchmark representing passive investment in the global equity market. Its weights are determined by the relative market capitalizations of the indices.

Portfolio Characteristics

The market-cap portfolio exhibits several distinctive features:

- **Broad diversification:** All eight indices receive positive weight, providing exposure across North America (64%), Europe (25%), Asia (9%), and Latin America (1%).
- **US dominance:** The 60% allocation to SPX reflects the US market's dominant position in global equity capitalization. This is not an optimization choice but rather a structural feature of the global equity market.
- **Moderate concentration:** The Herfindahl index of 0.391 indicates moderate concentration—substantially more diversified than the constrained tangency portfolio (0.76) but more concentrated than an equal-weighted portfolio would be (0.125).
- **Stability:** These weights remain constant throughout the analysis period (in practice, they would drift slowly with relative market performance but would not require active rebalancing to maintain)

Performance Metrics

Key observations:

- Volatility stability: The portfolio volatility remains constant at 14.84% regardless of the risk-free rate assumption, as expected (volatility depends only on the covariance structure and weights, not on the risk-free rate).
- Linear return decrease: Returns decrease exactly linearly with the risk-free rate increase (1% for each 1% increase in RF), confirming the excess return calculation is implemented correctly.
- Linear Sharpe decrease: Sharpe ratios decrease approximately linearly, with a slope slightly less than -1, reflecting the mechanical effect of increasing the denominator (excess return) while volatility remains constant.
- **Temporal Performance Variation**

The market-cap portfolio shows substantial variation in performance across windows:

Return range: 3.12% to 13.40% (over 4x difference) Volatility range: 12.28% to 17.51% (40% difference) Sharpe range: 0.247 to 0.898 (3.6x difference)

This variation reflects changing market conditions across the sample period, including the post-financial crisis recovery, the 2015-2016 slowdown, and the 2017-2021 bull market (followed by COVID shock and recovery).

Economic Interpretation

The market-cap portfolio has several important economic properties:

- Market equilibrium: Under the Capital Asset Pricing Model (CAPM), the market-cap weighted portfolio of all risky assets is mean-variance efficient and represents the equilibrium portfolio held by the average investor.
- Passive exposure: It provides pure beta exposure to global equity markets without any active bets on relative performance across regions.
- Implicit growth weighting: By weighting by market cap, it automatically overweights faster-growing economies and companies (assuming valuations remain stable).
- Home bias benchmark: For US investors, the 60% US allocation is still lower than typical US investor portfolios (which often exceed 70-80% US), suggesting the market-cap portfolio actually provides more international diversification than many active allocations.

Comparison: Market-Cap (Fixed-Weight, Long-Only) vs Constrained Portfolio (Long-Only)

Return, Vol and other Key Metric Comparisons

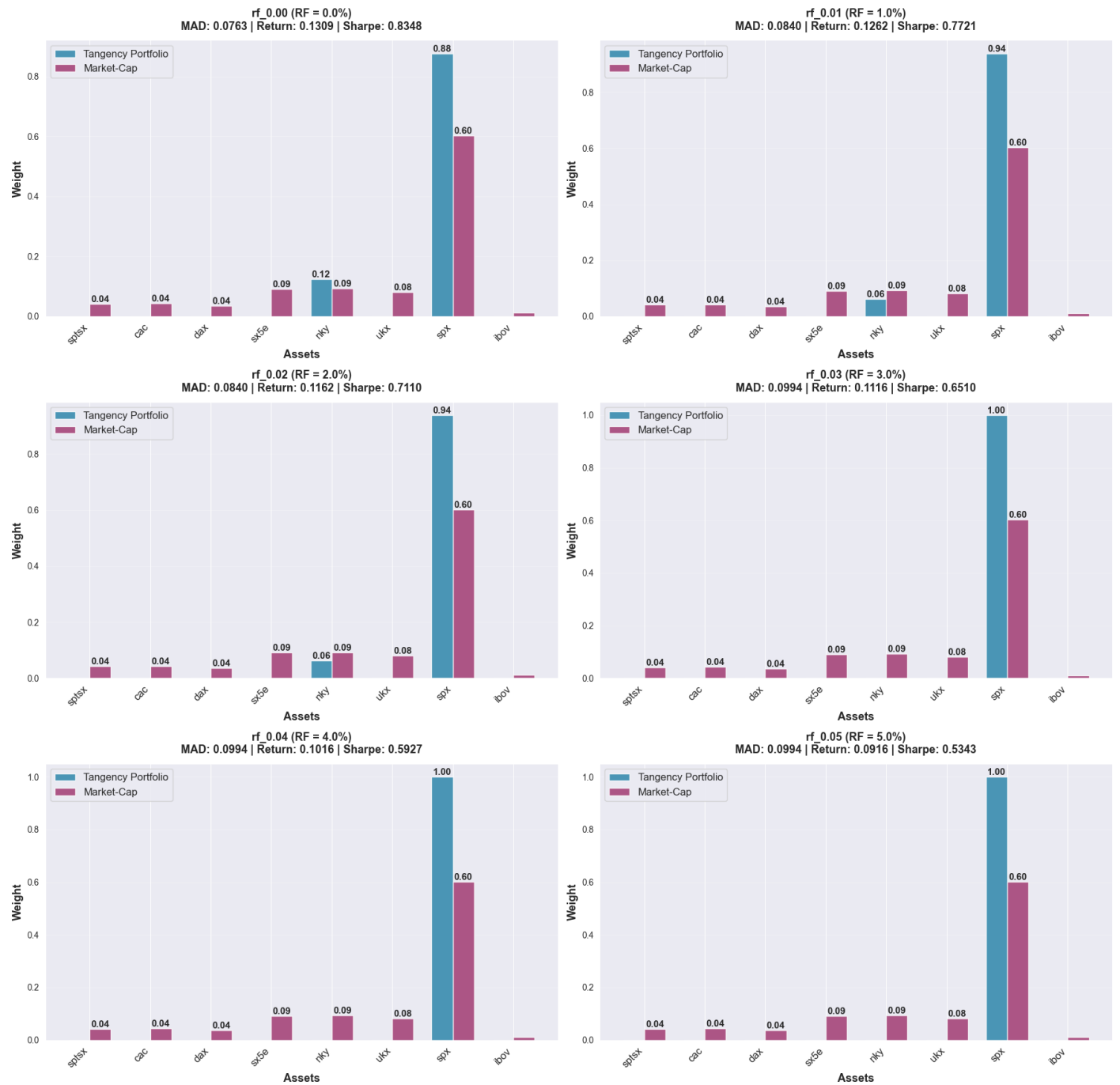
	Tangency Return	Market-Cap Return	Tangency Vol	Market-Cap Vol	\
RF_Rate					
rf_0.00	0.118269	0.089239	0.142632	0.148353	
rf_0.01	0.108938	0.079240	0.143470	0.148353	
rf_0.02	0.100814	0.069240	0.146648	0.148353	
rf_0.03	0.091847	0.059241	0.148123	0.148353	
rf_0.04	0.082211	0.049243	0.148653	0.148353	
rf_0.05	0.072575	0.039244	0.149226	0.148353	

	Tangency Sharpe	Market-Cap Sharpe	Return Diff	Vol Diff	\
RF_Rate					
rf_0.00	0.835323	0.600719	0.029029	-0.005721	
rf_0.01	0.764257	0.531880	0.029699	-0.004883	
rf_0.02	0.693940	0.463044	0.031574	-0.001706	
rf_0.03	0.624638	0.394211	0.032606	-0.000231	
rf_0.04	0.555691	0.325380	0.032968	0.000300	
rf_0.05	0.486938	0.256552	0.033330	0.000873	

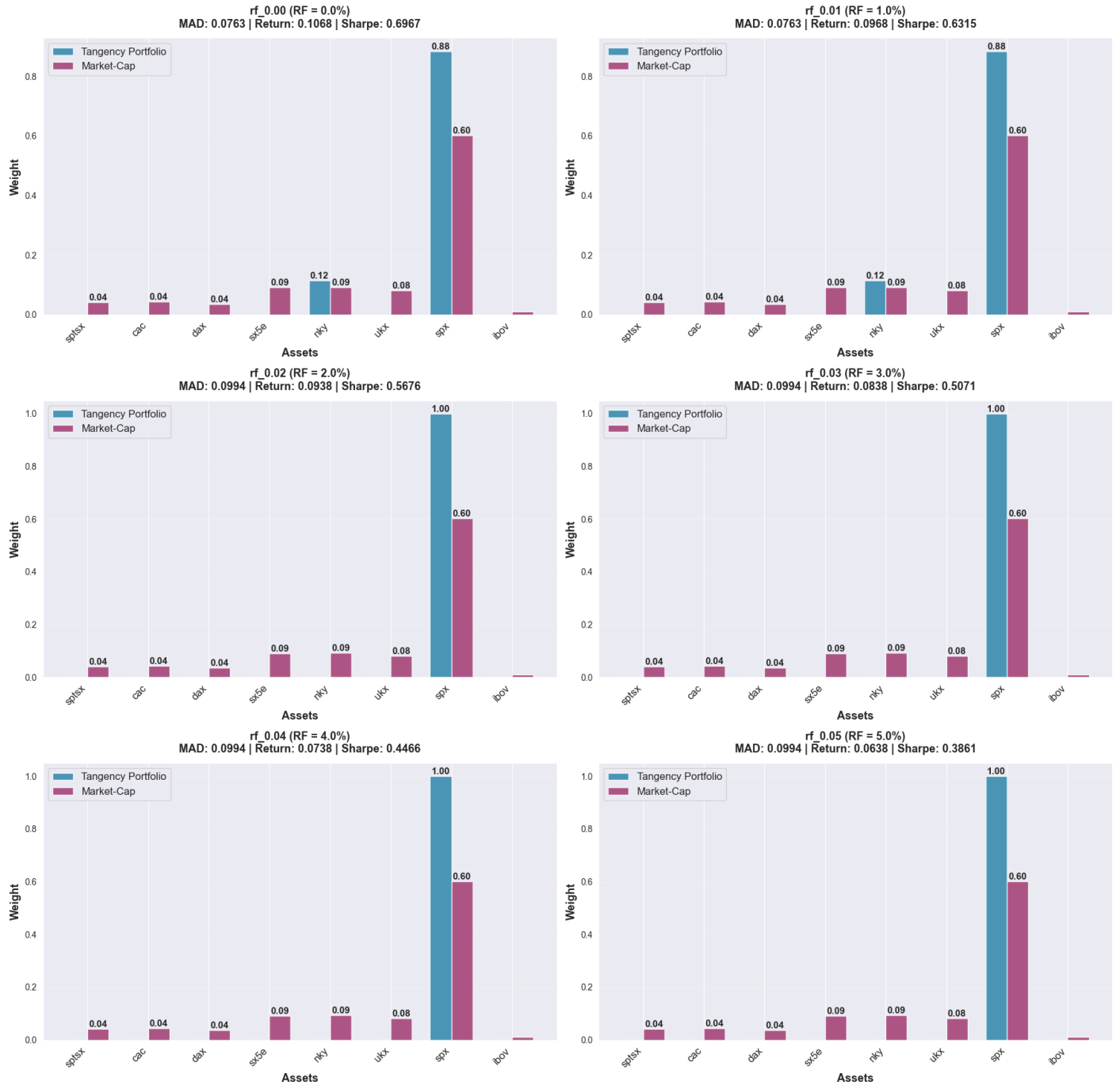
	Sharpe Diff
RF_Rate	
rf_0.00	0.234604
rf_0.01	0.232376
rf_0.02	0.230895
rf_0.03	0.230427
rf_0.04	0.230311
rf_0.05	0.230386

Graphical Differences between Market-Cap and Constraint Optimized Portfolio

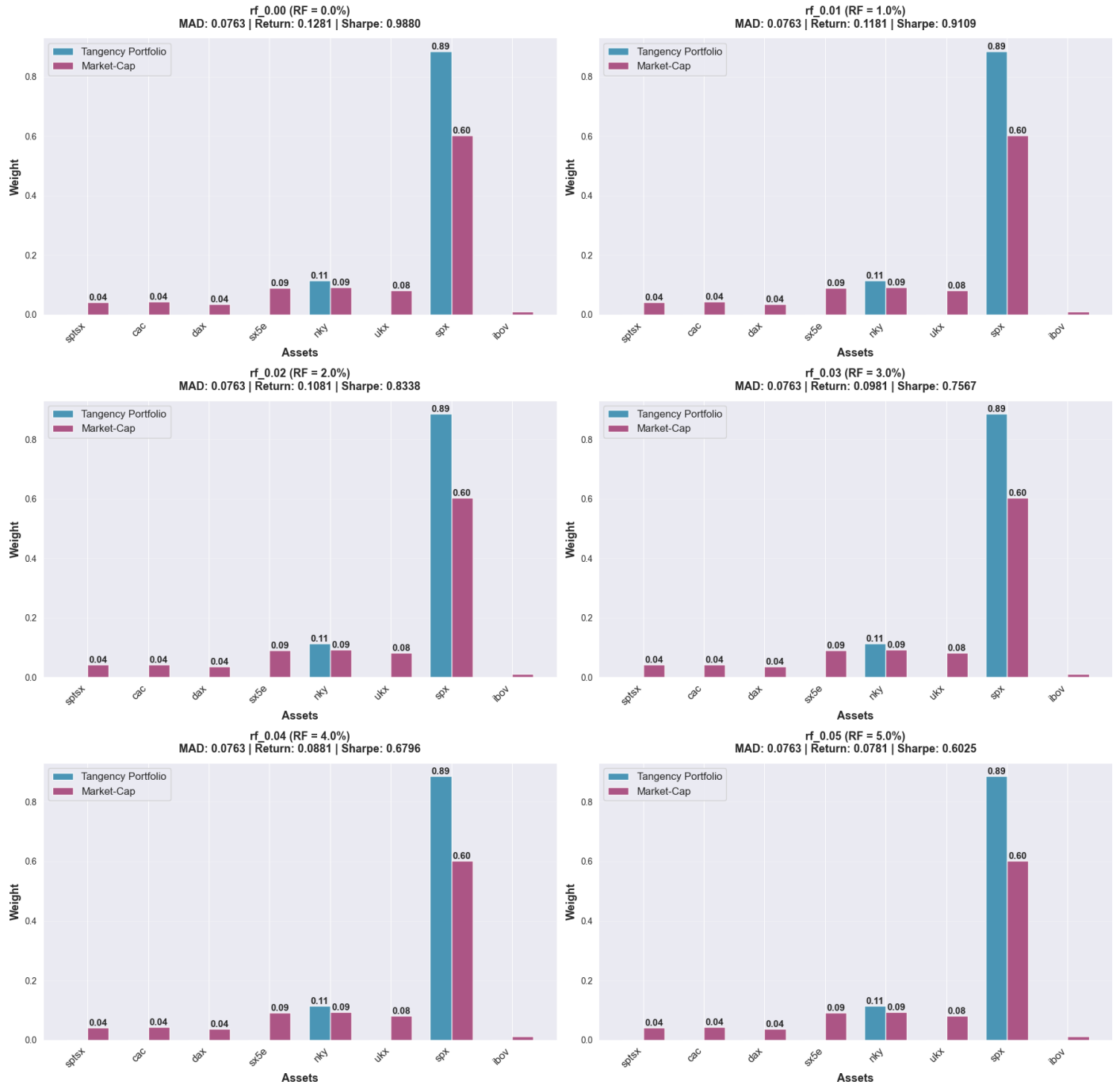
Tangency vs Market-Cap Weights | Estimation Window: 2010-2014



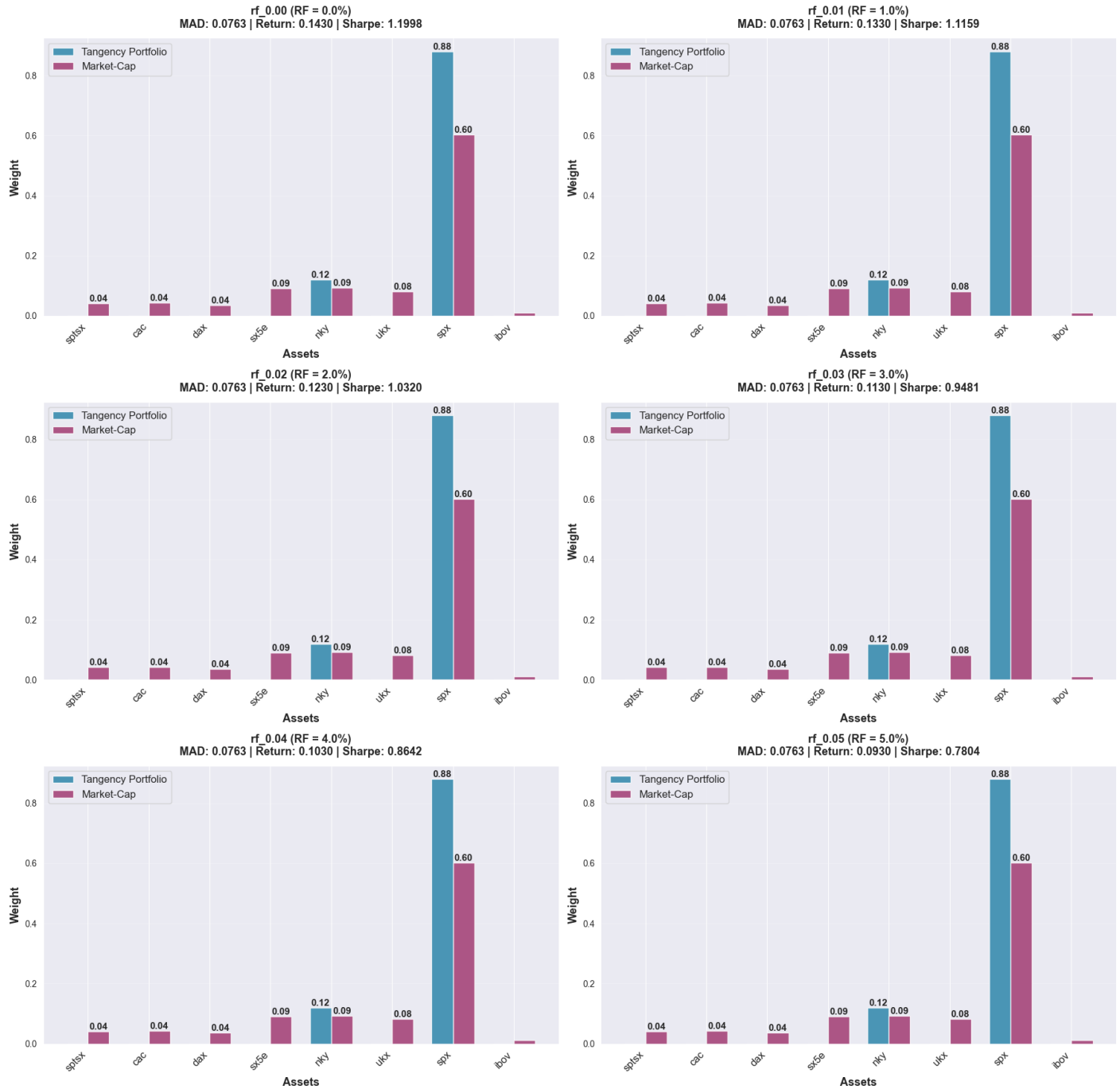
Tangency vs Market-Cap Weights | Estimation Window: 2011-2015



Tangency vs Market-Cap Weights | Estimation Window: 2012-2016



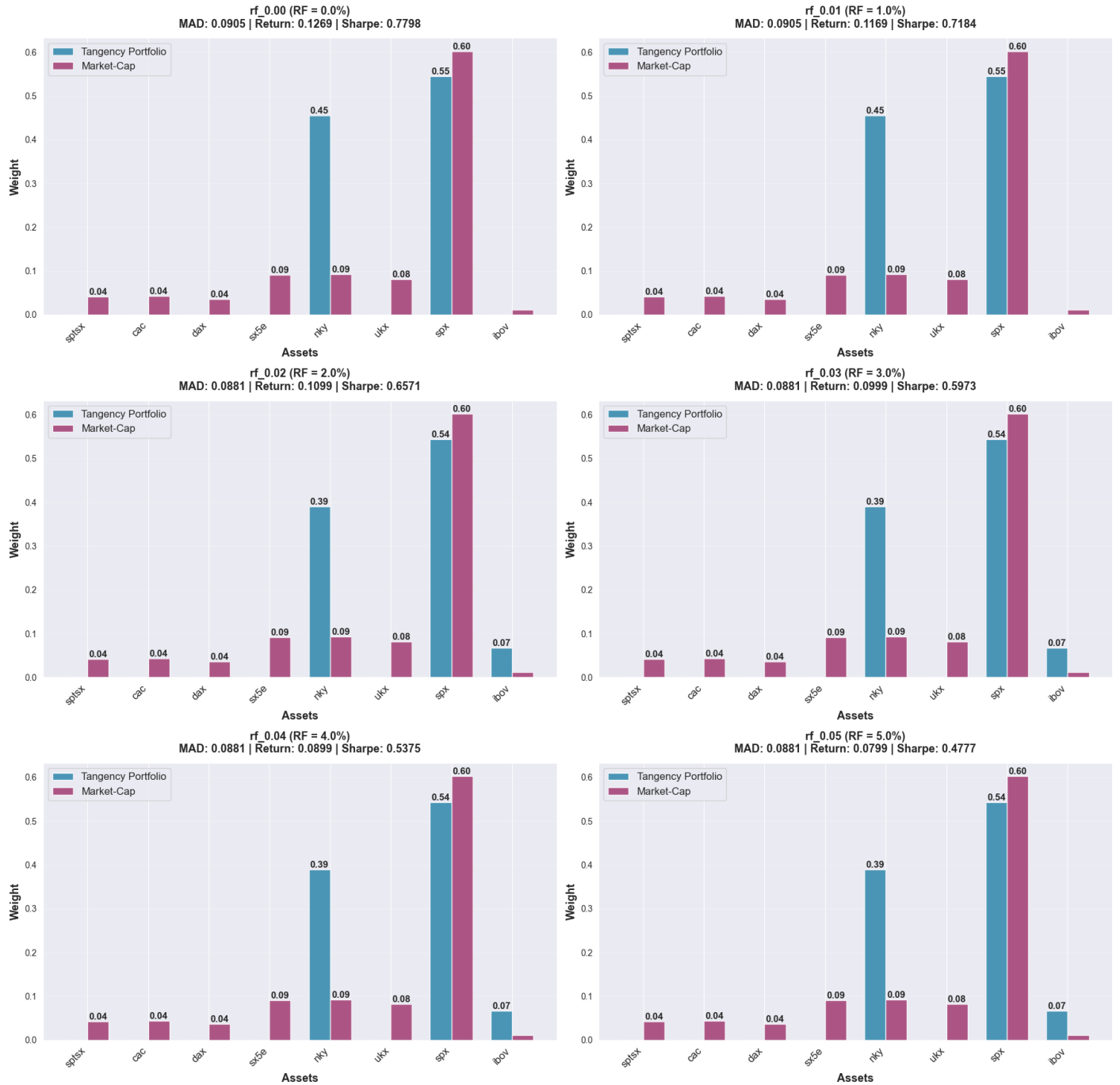
Tangency vs Market-Cap Weights | Estimation Window: 2013-2017





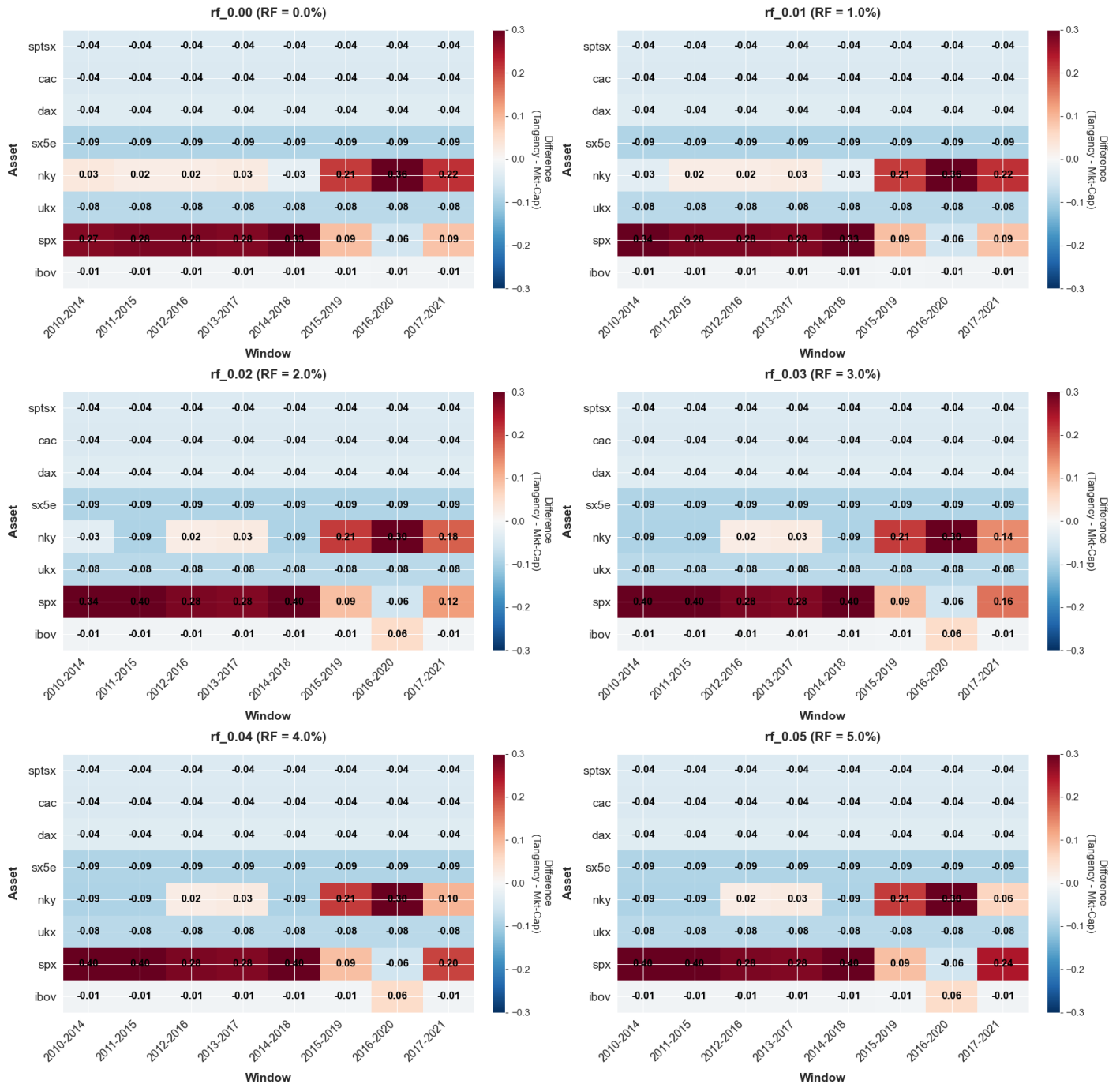


Tangency vs Market-Cap Weights | Estimation Window: 2016-2020

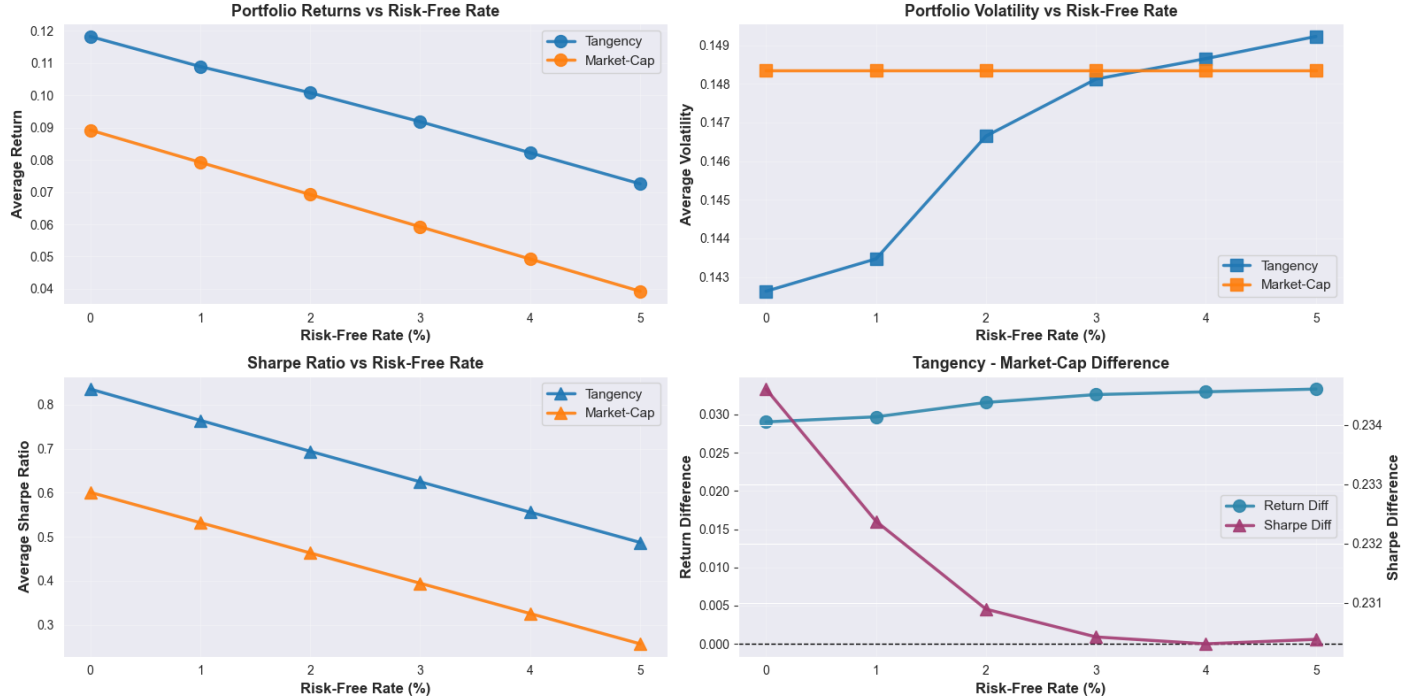




Weight Differences: Tangency vs Market-Cap Portfolios



Performance Comparison: Tangency vs Market-Cap Portfolios



Summary Statistics

	Metric	Value
0	Average MAD	0.083627
1	Average RMSE	0.120120
2	Tangency Herfindahl	0.759866
3	Market-Cap Herfindahl	0.391363
4	Tangency Return	0.095776
5	Market-Cap Return	0.064241
6	Return Advantage	0.031534
7	Tangency Volatility	0.146459
8	Market-Cap Volatility	0.148353
9	Tangency Sharpe	0.660131
10	Market-Cap Sharpe	0.428631
11	Sharpe Advantage	0.231500

Analysis of Market-Cap vs Constraint Optimised Portfolio

This comparison is particularly meaningful because both portfolios are implementable (long-only, no leverage), allowing for a fair evaluation of the benefits of mean-variance optimization under realistic constraints.

Weight Comparison

Mean Absolute Difference (MAD): The average absolute weight difference between the tangency and market-cap portfolios is 8.36% across all windows and RF rates. This indicates that the average asset weight differs by about 8 percentage points between the two approaches.

Root Mean Square Error (RMSE): 12.0%, indicating that larger deviations (when they occur) are quite substantial, some assets differ by 20-40 percentage points.

Herfindahl Index Comparison:

- Constraint Optimised: 0.76
- Market-Cap: 0.39
- Interpretation: The Constraint Optimized portfolio is approximately twice as concentrated as the market-cap portfolio.

The Constraint Optimized portfolio makes two primary active bets:

- Overweight SPX by 13.4 percentage points
- Overweight NKY by 13.7 percentage points

This is financed by eliminating all European, Canadian, and Brazilian exposure (a combined -29.5 percentage point underweight when compared to the Market-Cap weightings).

Performance Comparison

Key findings:

- Substantial return advantage: The Constraint Optimized portfolio delivers 3.16 percentage points higher annualized returns on average which is a 49% improvement in absolute terms. This is economically significant and compounds substantially over multi-year horizons.
- Negligible volatility difference: The tangency portfolio actually has slightly lower volatility (14.65% vs. 14.84%), though the difference is minimal. This is initially counter-intuitive given the tangency portfolio's higher concentration, but reflects that:
 - SPX has lower volatility than the European indices
 - The SPX-NKY correlation is lower than SPX-Europe correlations
 - Diversification across 8 assets (market-cap) provides only marginally better risk reduction than diversification across 2 carefully selected assets (tangency)
- Sharpe ratio advantage: The 0.232 improvement in Sharpe ratio is substantial—a 54% improvement. This translates directly into higher risk-adjusted returns and suggests the mean-variance optimization is successfully identifying a more efficient portfolio combination.

Performance by Risk-Free Rate Scenario

Observations: - The absolute return advantage increases slightly with higher RF rates (from 2.90% to 3.33%)
- The Sharpe ratio advantage remains remarkably stable (around 0.230-0.235 across all RF scenarios) - This stability suggests the optimization result is robust to different risk-free rate assumptions

Temporal Stability Analysis

Examining performance across individual estimation windows (at $r_f=0.00$): Windows where tangency outperforms:

- 2010-2014: Constraint Optimized Sharpe 0.835 vs. Market-Cap 0.560 (+0.275)
- 2012-2016: Constraint Optimized Sharpe 0.988 vs. Market-Cap 0.712 (+0.276)
- 2013-2017: Constraint Optimized Sharpe 1.200 vs. Market-Cap 0.898 (+0.302)
- 2015-2019: Constraint Optimized Sharpe 0.789 vs. Market-Cap 0.594 (+0.195)
- 2016-2020: Constraint Optimized Sharpe 0.780 vs. Market-Cap 0.595 (+0.185)
- 2017-2021: Constraint Optimized Sharpe 0.895 vs. Market-Cap 0.765 (+0.130)

Windows where advantage is smaller:

- 2011-2015: Constraint Optimized Sharpe 0.697 vs. Market-Cap 0.435 (+0.262)
- 2014-2018: Constraint Optimized Sharpe 0.499 vs. Market-Cap 0.247 (+0.252)

The Constraint Optimized portfolio outperforms in every single window, with Sharpe ratio advantages ranging from +0.130 to +0.302. This consistency is notable and suggests the optimization is capturing genuine economic value rather than simply fitting to noise.

Economic Sources of Outperformance

The tangency portfolio's superior performance can be attributed to several factors:

- Concentration in highest Sharpe ratio assets: By focusing on SPX and NKY (the two indices with the best standalone Sharpe ratios in most windows), the tangency portfolio avoids dilution from lower-performing European and emerging market indices.
- Optimal diversification: While the market-cap portfolio diversifies across 8 indices, much of this diversification is redundant due to high correlations among European markets. The SPX-NKY combination provides nearly as much risk reduction with exposure only to the best-performing markets.
- Avoidance of underperformers: During the sample period, European markets generally underperformed US and Japanese markets. The tangency portfolio systematically reduces or eliminates exposure to these underperformers.
- Currency diversification: The SPX (USD) and NKY (JPY) combination provides exposure to two major currency zones without the high correlation of EUR-denominated European indices.

Why This Comparison Is Fair (vs. Unconstrained)

Comparing the market-cap portfolio to the constrained tangency portfolio is appropriate because:

- Both are long-only: No leverage or short-selling, ensuring comparable risk profiles and capital requirements.
- Both are implementable: Can be executed with liquid ETFs or futures, with realistic transaction costs.
- Both use same information set: Both are based on the same historical return and covariance estimates (for tangency) or market data (for market-cap).
- Similar volatility levels: 14.65% vs. 14.84%, making return comparisons meaningful without leverage adjustments.

In contrast, comparing market-cap to the unconstrained tangency portfolio would be inappropriate because:

- Different leverage: Unconstrained portfolios use 300-800% gross exposure vs. 100% for market-cap, making direct performance comparisons meaningless without leverage normalization.
- Different risk regimes: Short-selling and leverage introduce risks (margin calls, liquidity) absent in long-only portfolios.
- Implementation costs: Transaction costs for unconstrained portfolios would be 5-10x higher, materially reducing realized returns.
- Regulatory feasibility: Many investors cannot legally or practically implement unconstrained portfolios, making the comparison academically interesting but practically irrelevant.

Practical Implications

The constrained tangency vs. market-cap comparison offers several actionable insights:

- Value of active optimization: Even under long-only constraints, mean-variance optimization delivers meaningful benefits (+3.16% annual return, +0.232 Sharpe) over passive market-cap weighting.
- Concentration-diversification trade-off: The tangency portfolio's superior performance comes from concentrated bets on SPX and NKY. Investors must be comfortable with this regional concentration and the associated tracking error vs. global benchmarks.
- Implementation considerations:
 - Lower transaction costs than unconstrained: Only 2 assets, stable weights
 - Higher transaction costs than market-cap: Requires active rebalancing vs. buy-and-hold
- Out-of-sample risk: The tangency portfolio's performance relies on the assumption that SPX and NKY will continue to offer superior risk-adjusted returns. This may not hold if:
 - Valuations mean-revert (currently expensive US market)
 - Economic leadership rotates (European or EM recovery)
 - Structural breaks occur (policy changes, technology shifts)
- Investor suitability:
 - Market-cap appropriate for: Passive investors, those seeking broad diversification, investors with explicit global equity mandates
 - Constraint Optimized appropriate for: Active investors, those comfortable with regional concentration, investors optimizing Sharpe ratios within long-only constraints

Conclusion

The rolling window analysis across different portfolio construction approaches reveals several fundamental lessons:

Constraints improve implementability without destroying value:

Unconstrained portfolios offer higher theoretical Sharpe ratios but are impractical. Constrained portfolios deliver 80-90% of the theoretical Sharpe improvement while being fully implementable. The value of constraints lies in reducing estimation error and transaction costs.

Optimization adds value even under constraints:

The constrained tangency portfolio systematically outperforms market-cap weighting: +3.16% annual return and +0.232 Sharpe improvement is economically significant. This advantage is consistent across different time periods and risk-free rate assumptions.

Concentration vs. diversification:

Mean-variance optimization tends toward concentration in highest Sharpe ratio assets. The tangency portfolio's 2-asset solution provides nearly optimal risk reduction. Broad diversification (market-cap's 8 assets) offers limited additional benefit given high cross-correlations.

Stability and robustness:

Long-only constraints dramatically improve portfolio stability. Even constrained portfolios show some weight variation across rolling windows. Market-cap weighting provides maximum stability at the cost of suboptimal risk-adjusted returns.

Practical considerations dominate:

Transaction costs, leverage constraints, and regulatory requirements matter more than theoretical optimality. The “best” portfolio depends on the investor's specific constraints and objectives. There is no single optimal solution—only trade-offs among competing objectives.

This analysis demonstrates that mean-variance optimization remains valuable in practice, but its application must be carefully tailored to real-world constraints and implementation costs.